

REVIEW ARTICLE



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Drug information-seeking behaviours of physicians, nurses and pharmacists: A systematic literature review

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Abstract

Background: Medication use typically involves physicians prescribing, pharmacists reviewing, and nurses administering medications to patients. Drug information (DI) is often required during the process, with the various health care professionals (HCPs) seeking information differently according to their needs and familiarity with various resources.

Objective: This systematic literature review aims to evaluate studies on drug information-seeking behaviour (ISB) of physicians, nurses and pharmacists to ascertain their DI needs, DI sources used, facilitators and barriers to DI-seeking.

Methods: A systematic search was conducted on PubMed, Embase.com, SCOPUS, PsycINFO, CINAHL and Cochrane Library to identify eligible primary research articles published between January 2000 and May 2020.

Results: The reviewed studies ($N = 48$) revealed that HCPs have a wide range of DI needs, with the top needs being similar across the three HCPs. Information sources used most often by all three groups were tertiary, followed by human and primary sources. Factors relating to the source characteristics were the most reported facilitators and barriers to DI-seeking. Some differences in drug ISB were also identified.

Conclusion: Our findings can also guide information providers and educators to optimize information provision. It may also facilitate effective communication amongst HCPs when obtaining DI from or providing DI to one another.

KEYWORDS

doctors; information needs analysis; information seeking behaviour; information sources; nurses; pharmacists; review, systematised

INTRODUCTION

Medication use is a process that involves various health care professionals (HCPs) in the provision of patient care. It is an interprofessional practice that typically involves the physicians prescribing the medications, pharmacists reviewing the orders, and nurses administering the

medications to patients. In this process, drug information (DI) may be required to ensure safe and effective use of medications for patients. Indeed, current literature reports that DI is one of the most commonly needed information by physicians, nurses and pharmacists (Clarke et al., 2013; Kostagiolas et al., 2011). The information could be related to dosages, drug interactions, use

in special populations, re-constitution or compounding, availability and others. The resources used to obtain such information may include drug compendia, databases, DI pharmacists, sales representative, the Internet and others.

While drug use is common to all three HCPs, the types of DI required or resources used by the various HCPs may be different due to the nature of their work. Past reviews on the information-seeking behaviour (ISB) of physicians, nurses and dentists have revealed similarities and differences in the information sources used. Physicians, nurses and dentists rely heavily on their colleagues for information, particularly specialists and more experienced colleagues (Clarke et al., 2013; Isham et al., 2016; Younger, 2010). A recent study on doctors also revealed that consultation with colleagues, journal articles, Internet websites, textbooks, and MEDLINE/PUBMED were sequentially the most used information sources (Daei et al., 2020). Meanwhile, hospital nurses were found to prefer Google and peers more than bibliographic databases (Alving et al., 2018). With regards to information needs, physicians sought information relating to treatment and diagnosis (Daei et al., 2020), while nurses' information needs were regarding protocols and procedures (Clarke et al., 2013).

However, these reviews were typically on general medical or health information, and not specific to information pertaining to drugs (Alving et al., 2018; Clarke et al., 2013; Daei et al., 2020; Younger, 2010). Since drug use is a complex process (Shalini et al., 2010), the ISB for DI likely differs from that of general health information and we hypothesize that there exists some similarities and differences in the drug ISB of different HCPs.

ISB as defined by Wilson (2000) is the '*purposive seeking for information as a consequence of a need to satisfy some goal. In the course of seeking, the individual may interact with manual information systems (such as a newspaper or a library), or with computer-based systems (such as the World Wide Web)*'. By adopting Wilson's (2000) definition of ISB to the DI context, this review defines drug ISB as the purposive seeking for DI due to a need to satisfy a *health care-related* goal, in which the individual may engage *DI systems or sources* while seeking. Following this definition, DI needs and DI sources were chosen as the data to be extracted and analysed for this systematic literature review. In addition, data on facilitators and barriers to DI-seeking was extracted to further explore the factors influencing the HCPs' choice of DI needs and sources.

In view of the sparse evidence on ISB specific to DI for each profession, systematic reviews can prove beneficial in drawing generalizable, comprehensive and accurate conclusions (Gopalakrishnan & Ganeshkumar, 2013) valuable for information providers and educators. Hence, this study aims to systematically review studies on drug ISB of

Key Messages

- Physicians may require more information about drug costs, whereas pharmacists are more concerned about advising patients.
- Source credibility and ease of access are the two key facilitators (or barriers) to obtain DI.
- More evidence is required on the drug information seeking behaviour of nurses.
- Information providers should aim to increase the ease of access to credible drug information sources.
- Health care educators should aim to tailor their programs to the DI needs of the health care professionals they are targeting.

physicians, nurses and pharmacists, to ascertain (1) their DI needs, (2) DI sources used and (3) facilitators and barriers to DI-seeking. Our findings will allow the different HCPs to appreciate and understand one another's perspectives on DI needs which may facilitate effective communication when obtaining information from or providing information to one another. Additionally, given today's intimidating amount of DI and resources available which may pose a challenge in locating the required information (Davies & Harrison, 2007) in a timely manner, information providers can tailor specific information according to the different HCPs' ISB, which may ultimately improve patient care, as it has been found that the inability to access specific DI promptly may result in an increase in adverse drug events and hence, unnecessary health care costs (Bates et al., 1997).

METHODOLOGY

This systematic literature review was conducted in-line with Butler et al.'s (2016) guide for qualitative systematic review and reporting was done in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). The PRISMA 2020 Checklist is presented in Table A1.

Search strategy

Six databases relevant to the topic were identified: PUBMED, CINAHL Plus with Full Text (via EBSCOhost), Embase.com, SCOPUS, PSYINFO (via Ovid) and COCHRANE LIBRARY (via Wiley). The Population, Context and Outcome (PCO)

framework (Butler et al., 2016) designed for qualitative systematic reviews, was then used to identify the following keywords: 'Physicians'(P), 'Nurses'(P), 'Pharmacists'(P), 'Drug Information'(C), 'Information-Seeking Behaviour'(O), 'Information Needs'(O), 'Information Sources'(O) and 'Facilitators and Barriers to Information-Seeking'(O). After which, synonyms were developed and database-specific controlled vocabularies were identified. The terms were truncated where appropriate and combined using 'OR' and 'AND' Boolean operators. A first search conducted on 31st January 2019 was limited to articles from 1st January 2000 to 31st January 2019. Subsequently, the search was updated and re-ran again on 16th May 2020 to include articles from 1st February 2019 to 16th May 2020. In the search, limits were applied where possible to obtain English articles. Other than database searching, hand-searching was done on the reference lists of relevant review articles and a search was also conducted on Google Scholar. The final list of citations was then imported into EndNote X9 (Clarivate Analytics, USA) and screened for duplicates. The search strategy for the database search on 31st January 2019 and 16th May 2020 is presented in Tables B1–B12.

Eligibility criteria

This review included primary research studies employing qualitative, quantitative or mixed-methods, that reported on DI needs, DI sources, facilitators and/or barriers to DI-seeking. The study population had to include physicians, nurses and/or pharmacists.

To ensure selected studies reflect current practice, behaviours and advancement in health information technology, only studies published between 1st January 2000 and 16th May 2020 were included.

Furthermore, reviews, letters, short reports, editorials, comments, discussion papers, conference extracts and other grey literature were excluded. Studies that were too specific in context such as particular disease, drug class, or on students or specialists, were also excluded to ensure generalizability of the results.

Study selection

Following Kuske et al.'s (2017) selection process, retrieved articles underwent a two-stage screening process carried out by two independent investigators, based on the pre-defined eligibility criteria.

Firstly, title and abstracts were screened and categorized into 'included', 'unclear' or 'excluded'. Subsequently, articles identified as 'included' and 'unclear' were screened as full text for final inclusion. A third

reviewer was consulted if there was any disagreement unresolvable by discussion between the two reviewers.

Data extraction

The data extraction and analyses were done by two investigators. A data extraction form was developed, piloted on four articles and modified accordingly. All data for DI sources and needs reported in the 'Results' section and all data for barriers and facilitators reported in the 'Results' and 'Discussion' sections were extracted.

Data analysis

The data extracted were analysed for the three HCPs separately and as a whole, using a content analysis approach which *'seeks to quantify content in terms of predetermined categories and in a systematic and replicable manner'* (Bryman, 2016). It is a well-developed and widely used method in the social sciences (Dixon-Woods et al., 2005) which have been utilized by other authors exploring information behaviours in health care (Adams et al., 2009; Biernatzki et al., 2018; Julien & Fourie, 2015; Tlach et al., 2015). The content analysis method used by Tlach et al. (2015) was adapted for this study. Quality assessment was not performed for this systematic literature review.

Drug information needs and sources

To better reflect the needs and sources that participants viewed as important, the analysis included data ranked as top 10 in terms of frequencies or percentages, or data reported as the 'main' or 'top'. In addition, to reflect general behaviour, extracted data on DI sources that were specific to certain DI contexts (e.g., drug interactions) or type of source (e.g., books) were excluded in the analysis. Data were then inductively coded, following which similarities and differences among the codes were identified to establish new categories and subcategories.

Lastly, percentages were calculated using the frequency of articles reporting the respective categories and subcategories divided by the sum of articles included in the analysis for each type of DI needs or DI sources. The percentages refer to the individual rows. As articles may include more than one type of DI needs or DI sources, percentages within each column should not be expected to sum up to 100%. For illustration purposes, an example is shown below.

Example on calculation of percentage:

Total number of articles reporting DI needs = 5

Types of DI needs	Articles which report this DI need	% of articles (calculation)
A	Article #1, #2, #3	$\frac{3}{5} \times 100\% = 60\%$
B	Article #2, #4, #5	$\frac{3}{5} \times 100\% = 60\%$
C	Article #1	$\frac{1}{5} \times 100\% = 20\%$

Facilitators and barriers

The content analysis for facilitators and barriers included all data extracted, regardless of rankings. Similar to the analysis for needs and sources, the data were also inductively coded and compared to identify new themes and subthemes, and percentages were calculated in a similar manner.

RESULTS

Study selection

The literature search identified 2087 potential articles, in which 48 met the inclusion criteria. The study selection process is presented in Figure 1.

Characteristics of studies

The 48 included studies were conducted in Asia ($n = 16$, 33.3%), Europe ($n = 13$, 27.1%), North America ($n = 9$, 18.8%), Africa ($n = 5$, 10.4%), Australia ($n = 4$, 8.3%) and South America ($n = 1$, 2.1%). The populations studied were physicians ($n = 19$, 39.6%), nurses ($n = 4$, 8.3%), pharmacists ($n = 19$, 39.6%) or a combination ($n = 6$, 12.5%). A majority of the studies employed a quantitative ($n = 26$, 54.2%) or mixed-methods ($n = 18$, 37.5%) design and 36 (75%) collected data through questionnaires.

The total number of participants enrolled across the included studies was 12,264 (physicians: 8846; pharmacists: 2639; nurses: 779). An overview of the characteristics of the included studies is outlined in Table 1.

Drug information needs

The extracted data on DI needs from 16 studies went through content analysis, including two studies that had two unique participant groups (Layton et al., 2007; Theodorou et al., 2009) in which their data were taken as two separate studies. None of the articles studied nurses solely, therefore no results were presented for nurses. The analysis revealed a total of 26 DI categories (Table 2).

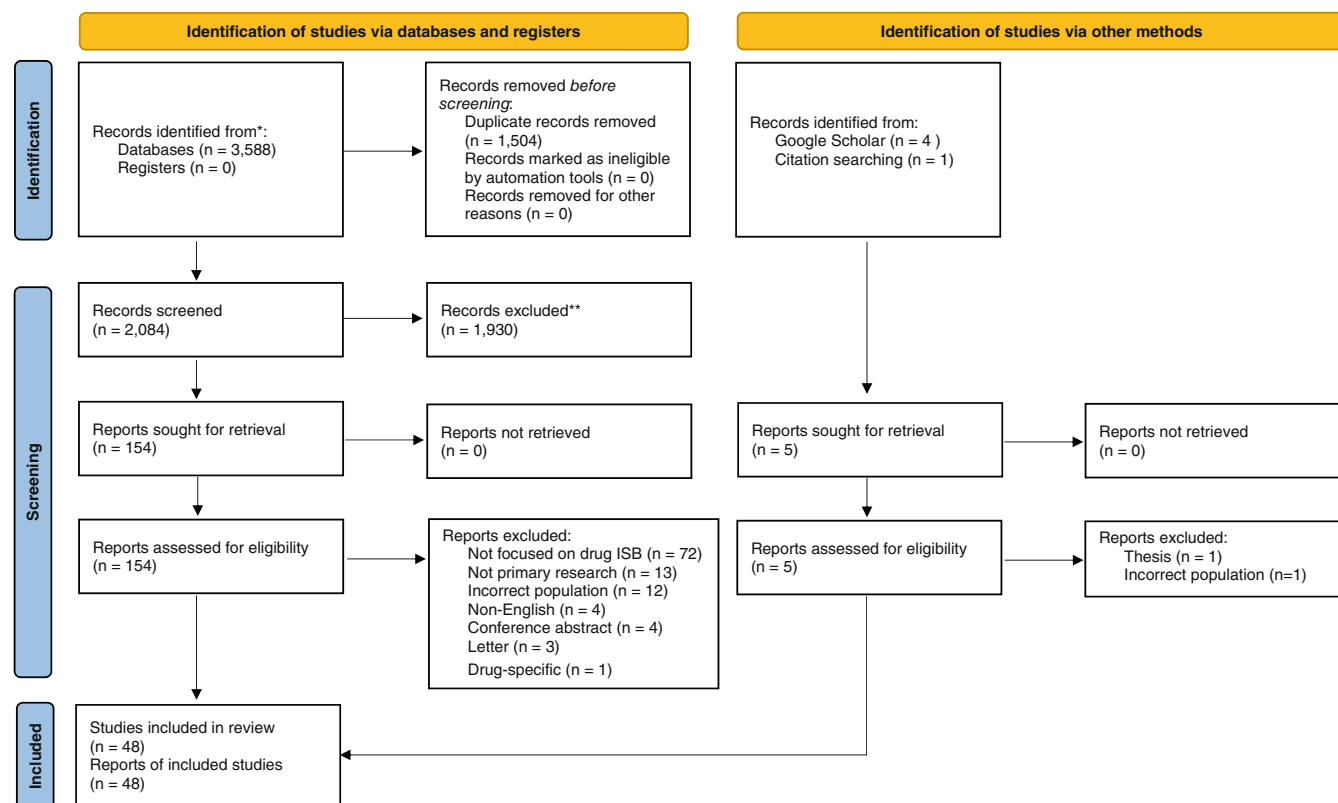


FIGURE 1 PRISMA 2020 flow diagram [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/hir.12472)]

TABLE 1 Summary characteristics of included studies

Reference	Method of data collection	Country	Recruitment setting: (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (% if available)	Tasks	
Quantitative studies							
Nabovati et al. (2017)	Questionnaire (method of administration unspecified)	Iran	Eight academic hospitals of MUMS; 86%; 315	281	Physicians Medical Residents (100%)	General	Current DDI information sources used by medical residents
Sharma et al. (2017)	Questionnaire (online or self-administered)	India	Clinicians in India's geographically dispersed areas; 83%; 220	183	Physicians Residents (72.1%); Specialist/consultant (27.9%)	General	Medicines information sources accessed by clinicians
Sharma et al. (2012)	Questionnaire (telephone or in-person interview)	India	Random, nationally representative sample of general practitioners and hospital doctors; N/A ^a ; N/A ^a	400	Physicians General Practitioners (50%); Hospital doctors (50%)	Prescribing	Prescribing information resources that Indian doctors use for new drugs, and whether hospital doctors and general practitioners differed in their use of the sources
Oshikoya et al. (2011)	Questionnaire (self-administered)	Nigeria	University College Hospital (UCH), Ibadan, Oyo State; 40.8%; 400	163	Physicians House officers (14.7%); Medical officers (13.5%); Junior residents (27.6%); Senior residents (29.4%); Consultants (14.7%)	Prescribing	Sources of DI for doctors working in a teaching hospital in Nigeria
Lua et al. (2011)	Questionnaire (online)	Singapore	Singapore Medical Association members who were general practitioners, medical officers, or house officers; 8.4%; 2821	236	Physicians Clinic-based physicians (49.2%); Hospital physicians (44.1%); Polyclinics physicians (6.8%)	General	Physicians' use of various sources for general DI
Theodorou et al. (2009)	Questionnaire (postal)	Greece & Cyprus	Proportional stratified sampling of physicians in Greece & Cyprus; 82.3% (Greece), 80.4% (Cyprus); 1463 (Greece), 240 (Cyprus)	1204 (Greece); 193 (Cyprus)	Physicians Work roles unspecified	Prescribing	Sources of physician information in Greece and Cyprus

(Continues)

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (% if available)	Tasks	
Tumwikirize et al. (2008)	Questionnaire (semi-structured self-administered)	Uganda	Public hospitals in Uganda, including six district hospitals, six regional and two university hospitals; 91%; 369	335	Physicians <i>Consultant (10.8%); Specialist (17.6%); Senior House Officer (7.3%); Medical Officer (36.0%); Intern (9.5%)</i>	General	Medicines information sources accessed by physicians in public hospitals in Uganda
Bergk et al. (2004)	Questionnaire (postal)	Germany	Random selection of general practitioners registered in April 2003 for reimbursement by state health insurances in Baden-Württemberg; 60.8%; 2000	1216	Physicians <i>General Practitioners (100%)</i>	General	DDI information required by General Practitioners with respect to both content and mode of presentation
Matsui et al. (2003)	Questionnaire (postal)	Canada	Physicians selected from the Canadian Medical Directory, 2001 edition; 52%; 500	261	Physicians <i>Family Physicians (100%)</i>	General	Where family physicians obtain prescribing information
Ponampalam and Anantharaman (2003)	Questionnaire (postal)	Singapore	Singapore Medical Council's (SMC) lists of registered medical practitioners for 1997; 24%; 4465	1071	Physicians <i>Clinic-based GP (33.2%); Hospital-based GP (26.1%); Private Specialist (14.3%); Institution-based Specialist (23.2%); Unknown (3.2%)</i>	General	Current information resources utilized by Singapore physicians and needs for drug and poisons information
McGettigan et al. (2001)	Questionnaire (postal)	Ireland	GPs selected from the national register; Hospital doctors from three hospitals affiliated with the Trinity College medical school; 54% (GP), 51% (Hospital doctors); 200 (GP), 230 (Hospital doctors)	226	Physicians <i>General Practitioners (47.8%); Hospital doctors (52.2%)</i>	Prescribing	Sources used in practice by doctors and whether hospital and primary care physicians differed in their use of the sources

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (%; if available)	Tasks	
Buckley et al. (2015)	Questionnaire (online)	Australia	Australian NPs who are members of Australian College of Nurse Practitioners (ACNP); 83%; N/A ^a	68 (Year 2007); 209 (Year 2010)	Nurses <i>Nurse Practitioners (100%)</i>	General	Sources currently used by Australian nurse practitioners to obtain information regarding quality use of medicines
Moorman et al. (2017)	Questionnaire (online)	USA	Pharmacists listed in the Department of Professional Licensing database; 15.2%; 1431	147	Pharmacists <i>Community Pharmacists (44%); Hospital Pharmacists (56%)</i>	General	Drug references Utah pharmacists use most frequently
Asmelashe Gelayee et al. (2017)	Questionnaire (structured interview)	Ethiopia	Community pharmacies in Gondar Town, Northwest Ethiopia; N/A ^a ; N/A ^a	48	Pharmacists <i>Community Pharmacists (100%)</i>	General	Needs and resources of DI in community pharmacies
Al-Farsi et al. (2014)	Questionnaire (self-administered)	Oman	Community pharmacies in different parts of Oman; 80%; N/A ^a	71	Pharmacists <i>Community Pharmacists (100%)</i>	General	Information resources of medicine used by the community pharmacists in various parts of Oman
Khan and Hadi (2014)	Questionnaire (online)	Saudi Arabia	Preceptors that had ever participated in APPE and IPPE modules offered by the college; 66%; 27	18	Pharmacists <i>Preceptors (100%)</i>	General	Use of electronic DI resources by clinical pharmacy preceptors engaged in APPE and IPPE clerkships
Carvajal et al. (2013)	Questionnaire (postal)	USA	Community and hospital pharmacists' names and addresses provided by KM Lists; 11.4%; 1000	114	Pharmacists <i>Community Pharmacists; Hospital Pharmacists</i>	General	Use of DI resources by pharmacists
Usir et al. (2012)	Questionnaire (postal)	Malaysia	Address list of pharmacies in Malaysia; 18.6%; 500	93	Pharmacists <i>Community Pharmacists (100%)</i>	General	Usage of printed and electronic references and Patient Medication Record in community pharmacy

(Continues)

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Tasks	Relevant study focus
				Number of participants	Population & work roles (%; if available)	Pharmacists		
Hassali et al. (2010)	Questionnaire (interview)	Malaysia	Convenience sample of community pharmacies at different districts of Penang Island; 78.3%; 83	65	Pharmacists (100%)	Community Pharmacists	Answering DI requests from patients and doctors	Types of DI resources used by community pharmacists in daily practice in Penang, Malaysia
Wong et al. (2009)	Questionnaire (online)	Singapore	Members of the Pharmaceutical Society of Singapore (PSS); 27.4%; 570	156	Pharmacists (51.3%); Hospital Pharmacists (48.7%)	Community Pharmacists	General	Where practicing pharmacists in Singapore usually obtain DI
Hennigen et al. (2009)	Questionnaire (postal)	Brazil	Drugstores and pharmacies placed at Metropolitan Region of Porto Alegre and regularly registered at the CRF-RS; 68.6%; 595	408	Pharmacists (Work roles unspecified)	Pharmacists	General	Use of DI sources by pharmacists in drugstores and pharmacies in southern Brazil
Udezi et al. (2007)	Questionnaire (method of administration unspecified)	Nigeria	Enugu, Nigeria; 68.2%; 85	58	Pharmacists (32%); Hospital Pharmacists (68%)	Community Pharmacists	General	Practice problems that require pharmacists to search for DI and the extent of utilization of the different sources of available DI
Schrimsher et al. (2006)	Questionnaire (postal)	USA	Alabama pharmacy facilities licensed by the Alabama State Board of Pharmacy; 42%; 1430	604	Pharmacists (79%); Hospital (12%); Others (6%); Home infusion (2%); Ambulatory care (0.2%)	Community Pharmacists	General	Usage of DI resources in Alabama pharmacy facilities
Raal et al. (2006)	Questionnaire (postal)	Europe	Database of subscribers of the Drug Information Bulletin; 23%; 2000	389	Physicians (82%) & Pharmacists (18%)	Physicians (82%) & Pharmacists (18%)	General	Current sources and perceived needs of DI among Estonian physicians and pharmacists

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (%; if available)	Tasks	
Adibe (2010)	Questionnaire (self-administered)	Nigeria	health facilities and Faculty of pharmaceutical sciences in Enugu state; 85.9%; 78	67	Physicians (41.8%) & Pharmacists (58.2%) <i>Medical doctors (41.8%); Pharmacists (58.2%)</i>	General	Behaviour of health professionals (physicians and pharmacists) in Enugu State, Nigeria towards provision of DI services in the state
Almazrou et al. (2019)	Questionnaire (online)	Saudi Arabia	Private and government hospitals throughout Saudi Arabia; 50.8%; 500	254	Physicians <i>Consultant (18.1%); Specialist (33.5%); Resident (48.4%)</i>	General	Ascertain information seeking behaviours and awareness of physicians regarding DIC services
Qualitative studies							
Norri-Sederholm et al. (2016)	Focus group	Finland	Workshop consisting of primary home care nurses representing different areas of Kuopio	15	Nurses <i>Primary home care Nurses (100%)</i>	Home care	Information needed and delivered by primary home care nurses in order to perform medication care successfully
Hall et al. (2003)	Interviews (semi-structured)	England	Five different health authorities in England	22	Nurses <i>Community Nurse Prescribers (100%)</i>	Prescribing	How nurses access pharmacological information
Baysari et al. (2013)	Interview (semi-structured); focus group	Australia	Teaching hospital in Sydney	35	Physicians (34%) & Nurses (43%) & Pharmacists (23%) <i>Work roles unspecified</i>	General	Electronic medication information needs of clinicians
Barker et al. (2019)	Interviews (semi-structured)	Canada	Community sample of community pharmacy managers within the Halifax Regional Municipality (Nova Scotia)	15	Pharmacists <i>Community pharmacy managers (100%)</i>	General	Explores the barriers that may limit the use of patient safety information sources within community pharmacies

(Continues)

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting: (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (%; if available)	Tasks	
Mixed-methods studies							
Hughes et al. (2015)	Questionnaire (online)	USA	Physician residents enrolled in the Department of General Internal Medicine of a community teaching hospital; 45.2%; 146	66	Physicians Medical residents (100%)	General	Medical residents' day-to-day use of DI resources
Rahmner et al. (2012)	Focus group; Questionnaire (method of administration unspecified)	Sweden	Physicians working in Stockholm metropolitan area; 100%; 18	18	Physicians Internal Medicine clinicians (22.2%); Geriatricians (38.9%); General practitioners (38.9%)	Prescribing	Perceived needs for DI among physicians in Sweden
Karapinar et al. (2010)	Questionnaire (postal)	Netherlands	St. Lucas Andreas Hospital in Amsterdam; 32%; 464	149	Physicians General practitioners (100%)	General	Needs of Dutch general practitioners on discharge medication, both regarding content, timing
Hedegaard and Damkier (2009)	Questionnaire (postal)	Denmark	Physicians who received a written problem-oriented answer related to a specific patient from the Drug Information Centre in Denmark; 93%; 197	183	Physicians General practitioners (53%); Secondary-care physicians (47%)	General	Kind of sources physicians search before applying for POD
Abou-Auda (2008)	Questionnaire (postal)	Saudi Arabia	Hospitals, health care centres, infirmaries (private or governmental) and private clinics throughout the KSA; 65.9%; 2000	1229 (415 GPs)	Physicians General Practitioners (33.8%); Consultants (56.5%); Dentists (9.8%)	General	Drug ISB of physicians in Saudi Arabia including information resources used to obtain information about new drugs and future information needs.
Thailand				Physicians		Prescribing	

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (% if available)	Tasks	
Layton et al. (2007)	Questionnaire (postal); Interviews (semi-structured)		Tertiary care public teaching hospital in a major city; 40%; 328	131	Faculty (44.3%); Residents (55.7%)		Sources and types of information about new drugs that Thai doctors at a teaching hospital perceived as important before prescribing
Goodwin et al. (2007)	Questionnaire (postal)	Canada	Family physicians in Ontario selected from the 2003 Canadian Medical Directory; 53%; 756	265	Physicians Family physicians (100%)	General	Sources of information of family physicians in providing information regarding pregnancy-related drug exposures
Ndosi and Newell (2010)	Questionnaire (method of administration unspecified)	England	Hospital in the North of England; 43%; 98	42	Nurses Staff nurses (81.0%); Charge nurses (16.7%); Enrolled nurse (2.4%)	Administering medicines	Sources of medicine information that nurses use while administering medicines
Borja-Hart and Leachman (2016)	Phone survey	USA	Licensed chain community pharmacies within the state of Tennessee that were listed on the Tennessee Department of Health website;	39	Pharmacists Community Pharmacists (100%)	General	Types of DI resources used by chain community pharmacists in Tennessee
Ming et al. (2016)	Questionnaire (online)	Malaysia	Pharmacists registered in 3 states in the central part of Malaysia: Selangor, Perak, and Negeri Sembilan; 49.8%; 450	224	Pharmacists Government hospital (68.9%); Pharmaceutical services divisions (11.6%); Health clinic (5.8%); Public university (4.5%); Community pharmacy (4%); National Pharmaceutical Control Bureau (3.6%); Private university (2.2%); Others (0.4%)	General	Use of HHCs among pharmacists to retrieve DI in their clinical practice

(Continues)

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (% if available)	Tasks	
Khan et al. (2013)	Questionnaire (face-to-face interviews)	Malaysia	Community pharmacists practicing in 3 populous areas (George Town, Gelugor, and Bukit Jambul) of Penang Island; 67.3%; 55	37	Pharmacists <i>Community Pharmacists (100%)</i>	General	Use of DI resources at a community pharmacy in Penang Island, Malaysia
Crank (2005)	Questionnaire (postal)	Australia	All NSW public hospital pharmacies, as published in the Directory of Hospital Pharmacy and Pharmaceutical Organizations; 57%; N/A ^a	150	Pharmacists <i>Hospital pharmacists (100%)</i>	General	Utilization of the NSW Medicines Information Centre and other DI services of pharmacists in NSW public hospitals
Zehnder et al. (2004)	Questionnaire (postal)	Switzerland	Community pharmacies from the German-speaking part of Switzerland; 48%; 223	108	Pharmacists <i>Community Pharmacists (100%)</i>	General	Types of DI used by pharmacists in daily practice, their needs and wishes concerning DI
Burgess et al. (2004)	Questionnaire (postal)	England	Community pharmacists located in South Manchester Primary Care Trust (PCT); 71%; 31	22	Pharmacists <i>Community pharmacists (100%)</i>	General	Medicines information needs of community pharmacists and where they currently obtain medicines information from
Jaradat and Sweileh (2003)	Questionnaire (postal)	Palestine	community pharmacies in Qalqilia, Nablus, Tulkaram, Jenin, Ramallah, Jerusalem and Hebron cities and rural areas; 79%; 150	118	Pharmacists <i>Community pharmacists (100%)</i>	General	Sources and needs of DI for community pharmacies in Palestine
Murphy et al. (2006)	Questionnaire (postal)	Canada	Licensed, actively practicing, primary health care NPs and their collaborating physician partners in Nova Scotia; 75%; 36	27	Physicians (48%) & Nurses (52%) <i>Physicians (48%); Nurse Practitioners (52%)</i>	General	DI resources used by NPs and their collaborating physician partners at the point of care

TABLE 1 (Continued)

Reference	Method of data collection	Country	Recruitment setting; (response rate; sample size—for questionnaires only)	Population characteristics			Relevant study focus
				Number of participants	Population & work roles (% if available)	Tasks	
Mallows et al. (2005)	Phone survey	Australia	Calls to the NSW PIC from doctors and nurses seeking poisoning management advice	498	Physicians (55.4%) & Nurses (44.6%) <i>Work roles unspecified</i>	General	Reference sources that physicians and nurses would utilize prior to calling the Poison Information Centre (PIC) and in the absence of access to a PIC
Rutter et al. (2004)	Questionnaire (postal)	UK	Southampton City, Eastleigh, and Test Valley South and The New Forest primary care trusts; 67%; 748	333	Physicians (44%) & Nurses (56%) <i>General practitioners (44%); Nurses (56%)</i>	General	Current utilization of the Wessex Regional Drug and Medicines Information Service by primary care users (general practitioners and nurses) and the alternate resources nonusers use to answer medicines-related questions

^aN/A: information is not available.

TABLE 2 Main or top 10 reported drug information needs of HCPs^a

Drug information needs			
Categories	Health care professionals ^b (N ^c = 16) % ^d	Physicians (N ^c = 9) % ^d	Pharmacists (N ^c = 4) % ^d
Dosage/administration	93.8 ^{e,f}	88.9 ^{e,f}	100
Adverse drug reactions	93.8 ^{e,f}	88.9 ^{e,f}	100
Drug interactions	68.8 ^{e,f}	77.8 ^{e,f}	50.0
Indications/unlabelled uses	56.3 ^e	33.3 ^e	75.0
Pregnancy & lactation	43.8 ^f	44.4 ^f	25.0
Contraindications	43.8 ^e	44.4 ^e	25.0
Compatibility and stability	25.0	22.2	50.0
Use in diseased states	25.0 ^f	22.2 ^f	0
Use in renal/liver disease	25.0 ^f	22.2 ^f	25.0
Pharmacokinetics/toxicokinetics	25.0	11.1	50.0
Toxicology/poison	18.8	22.2	25.0
Product availability	18.8	11.1	25.0
Pharmacology	18.8	0	25.0
Cost and economics	18.8 ^e	22.2 ^e	0
Efficacy	18.8 ^e	22.2 ^e	0
Identification	12.5	11.1	25.0
Use in elderly	12.5	0	0
Clinical trial results	12.5 ^e	22.2 ^e	0
Comparisons with other drugs	12.5 ^e	22.2 ^e	0
Complementary & alternative medicine	6.3	11.1	0
Drug overview	6.3	11.1	0
Therapeutics	6.3	11.1	0
Travel medicine	6.3	11.1	0
General drug information	6.3	11.1	0
Counselling	6.3	0	25.0
Generic/brand name	6.3	0	25.0

^aExtracted data included in the analysis comprised of: 'top 10' DI needs in terms of frequency/percentage, main/top DI needs, and most frequent DI needs.

^bHealth care professionals includes physicians, nurses and pharmacists.

^cNumber of articles, for each population, that reported on DI needs that was included in this analysis. Studies that involved two independent populations were counted as two studies.

^dPercentage of articles that reported on that category of DI need. This is calculated using the frequency of articles reporting the respective categories divided by the sum of articles included in the analysis for DI needs for that population (refer to c).

^eArticle (Layton et al., 2007) was counted as two studies due to the presence of two independent physician populations: faculty physicians and residents.

^fArticle (Theodorou et al., 2009) was counted as two studies due to the presence of two independent physician populations: physicians from Greece and physicians from Cyprus.

'Dosage/administration', 'Adverse drug reactions' and 'Drug interactions' were the top reported DI needs common across physicians, pharmacists and the combined group. Other top DI needs reported by four or more studies in the combined group were 'Indications/unlabeled uses', 'Pregnancy & lactation', 'Contraindications', 'Compatibility and Stability', 'Use in diseased states', 'Use in renal/liver disease' and 'Pharmacokinetics/toxicokinetics'.

Although less frequently reported, the DI needs which are identified for physicians but not pharmacists are 'Efficacy', 'Cost and economics', 'Comparisons with other drugs', 'Clinical trial results', 'Use in diseased states', 'Complementary & alternative medicine', 'Drug overview', 'Therapeutics', 'Travel Medicine' and 'General drug info'. Likewise, there were needs that pharmacists reported which were not for physicians. This included information

on 'Pharmacology', 'Counselling' and 'Generic/Brand name', which were each reported by one study.

There were further differences observed between the needs of physicians and pharmacists. Firstly, 'Pregnancy & lactation' (44.4%) and 'Contraindications' (44.4%) were within the top five reported needs for physicians but were only reported in one study (25.0%) for pharmacists. In addition, 'Indications/unlabeled uses' was more frequently reported for pharmacists (75.0%) compared to

physicians (33.3%). This was the same for 'Compatibility and stability' (50.0% versus 22.2%).

Drug information sources

The extracted data on DI sources from 29 studies underwent content analysis, including four studies that had two unique participant groups (Moorman et al., 2017;

TABLE 3 Main or top 10 reported drug information sources of HCPs^a

Drug information sources				
	Health care professionals ^b (N ^c = 29) % ^d	Physicians (N ^c = 10) % ^d	Nurses (N ^c = 4) % ^d	Pharmacists (N ^c = 13) % ^d
<i>Tertiary sources</i>	100	100	100	100
Reference books	82.8 ^{e,g,h}	70.0	100	84.6 ^e
Internet	51.7 ^{f,h}	70.0 ^f	75.0	30.8
Drug databases	51.7 ^{e,h}	30.0	75.0	69.2 ^e
Textbooks	37.9 ^f	50.0 ^f	25.0	38.5
Product-specific information	20.7 ^h	20.0	50.0	15.4
Pharmaceutical company materials	20.7	20.0	0	30.8
Guidelines	20.7 ^g	30.0	25.0	15.4
Dictionaries	10.3 ^e	0	0	15.4 ^e
Other organization materials	6.9	0	50.0	0
Online resources for HCPs	3.4	0	0	7.7
<i>Human sources</i>	82.8	100	100	61.5
Pharmaceutical companies/ representatives	58.6 ^{f,h}	80.0 ^f	50.0	38.5
Health care professionals	48.3 ^g	70.0	75.0	30.8
Professional activities	24.1 ^f	50.0 ^f	25.0	0
Information services	20.7 ^h	10.0	25.0	30.8
Own expertise	6.9 ^h	10.0	25.0	0
<i>Primary sources</i>	58.6	80.0	75.0	30.8
Journals	58.6 ^{f,g}	80.0 ^f	75.0	30.8
<i>Other sources</i>	20.7	30.0	25.0	0
Libraries	10.3 ^f	20.0 ^f	25.0	0
Governmental agency	6.9	0	0	0
Clinical information systems	6.9 ^h	10.0	25.0	0

^aExtracted data included in the analysis comprised of: 'top 10' DI sources in terms of frequency/percentage, main/top sources, and most frequently used DI sources.

^bHealth care professionals includes physicians, nurses and pharmacists.

^cNumber of articles, for each population, that reported on DI sources that was included in this analysis. Studies that involved two independent populations were counted as two studies.

^dPercentage of articles that reported on that category/subcategory of DI sources. This is calculated using the frequency of articles reporting the respective categories/subcategories divided by the sum of articles included in the analysis for DI sources for that population (refer to c).

^eArticle (Moorman et al., 2017) was counted as two studies due to the presence of two independent pharmacist populations: community pharmacists and hospital pharmacists.

^fArticle (Theodorou et al., 2009) was counted as two studies due to the presence of two independent physician populations: Greece physicians and Cyprus physicians.

^gArticle (Murphy et al., 2006) was counted as two studies due to the presence of two independent HCP populations: physicians and nurses.

^hArticle (Rutter et al., 2004) was counted as two studies due to the presence of two independent physician populations: faculty physicians and residents.

TABLE 4 Drug information-seeking facilitators reported by HCPs^a

Facilitators to drug information-seeking				
	Health care professionals ^b (N ^c = 26) % ^d	Physicians (N ^c = 10) % ^d	Nurses (N ^c = 3) % ^d	Pharmacists (N ^c = 12) % ^d
<i>Source characteristics</i>	92.3	80.0	100	100
Credible source (e.g., reliable, unbiased, high quality, updated information)	76.9	80.0	100	66.7
Ease of use	42.3	40.0	33.3	41.7
Speed of use	42.3	60.0	33.3	25.0
Adequate information in sources	34.6	20.0	66.7	33.3
Sources are frequently updated	26.9	10.0	33.3	41.7
Portability	23.1	10.0	66.7	16.7
Appropriate format/presentation	23.1	10.0	66.7	25.0
Presence of human interaction	7.7	10.0	0	0
Web-based	3.8	10.0	0	0
<i>Access characteristics</i>	73.1	60.0	66.7	83.3
Ease of accessing sources (including internet access)	38.5	30.0	33.3	41.7
Sources are readily available	26.9	20.0	0	41.7
Low or no cost	26.9	10.0	66.7	25.0
Timely access of sources	19.2	10.0	0	33.3
Access to technology (e.g., computers)	7.7	10.0	0	0
Sources are consolidated	7.7	10.0	0	0
<i>Individual characteristics</i>	34.6	40.0	0	33.3
Comfortable/familiar with sources	19.2	20.0	0	16.7
Trust/confidence in sources	7.7	10.0	0	8.3
Having critical appraisal skills	3.8	10.0	0	0
Knowledge regarding sources (i.e., availability, suitability, importance and content of sources)	3.8	0	0	8.3
<i>Environment characteristics</i>	7.7	0	33.3	0
Presence of peer support	7.7	0	33.3	0
<i>Others</i>	3.8	10.0	0	0
Information logistics	3.8	10.0	0	0

^aExtracted data included in the analysis comprised of all data reported that is related to facilitators to information-seeking, regardless of ranking.

^bHealth care professionals includes physicians, nurses and pharmacists.

^cNumber of articles, for each population, that reported on DI-seeking facilitators that was included in this analysis.

^dPercentage of articles that reported on that category/subcategory of facilitator. This is calculated using the frequency of articles reporting the respective categories/subcategories divided by the sum of articles included in the analysis for facilitators for that population (refer to c).

Murphy et al., 2006; Rutter et al., 2004; Theodorou et al., 2009) in which their data was taken as two studies each. A total of four main categories and 19 subcategories were derived inductively during the content analysis (Table 3).

Among the four main categories, tertiary sources were reported by all 29 studies (100%). This was followed by human sources (82.8%) and primary sources (58.6%).

HCPs mostly utilized tertiary sources like 'Reference books' (82.8%), 'Internet' (51.7%) and 'Drug databases' (51.7%). Reference books most cited were:

- *British National Formulary* (Adibe, 2010; Hall et al., 2003; Hassali et al., 2010; Ndosi & Newell, 2010; Rutter et al., 2004; Tumwikirize et al., 2008; Udezi et al., 2007; Wong et al., 2009),

TABLE 5 Drug information-seeking barriers reported by HCPs^a

Barriers to drug information-seeking				
	Health care professionals ^b (N ^c = 32) % ^d	Physicians (N ^c = 13) % ^d	Nurses (N ^c = 5) % ^d	Pharmacists (N ^c = 14) % ^d
<i>Source characteristics</i>	84.4	92.3	60.0	78.6
Issues with source credibility (e.g., unreliable, biased, low quality, outdated information)	56.3	69.2	40.0	50.0
Time-consuming usage	34.4	30.8	20.0	28.6
Inadequate information in sources	34.4	30.8	0	42.9
Information is not useful	15.6	15.4	20.0	21.4
Difficulty of use	15.6	15.4	0	21.4
Inconsistent information between sources	9.4	23.1	0	0
Too much information	9.4	7.7	20.0	7.1
Poor presentation (long text and not classified by clinical relevance)	9.4	15.4	40.0	0
Sources are less frequently updated	9.4	15.4	20.0	0
Poor portability	9.4	15.4	20.0	0
Sources need updating	6.3	7.7	0	0
Lacks human interaction	6.3	15.4	0	0
Not integrated with patient-specific information	6.3	15.4	0	0
Controversies in information	3.1	7.7	0	0
Lack of interest from human sources	3.1	0	20.0	0
Non-response from human sources	3.1	0	0	7.1
<i>Access characteristics</i>	68.8	53.9	60.0	71.4
Difficulties in accessing sources (including internet access)	40.6	30.8	40.0	42.9
Sources are unavailable (lacking, missing or have supply issues)	37.5	23.1	40.0	42.9
High or presence of cost	34.4	38.5	20.0	21.4
Lack of access to technology (e.g., computers)	9.4	7.7	0	14.3
Sources are not consolidated	9.4	15.4	0	7.1
Rules and regulations preventing access to sources	6.3	0	20.0	7.1
Unable to have timely access	3.1	0	20.0	0
Slow internet speed	3.1	0	0	0
<i>Individual characteristics</i>	65.6	38.5	60.0	92.9
Insufficient knowledge regarding sources (i.e., availability, suitability, importance and content of sources)	37.5	15.4	40.0	57.1
Lack of skills in searching sources	28.1	7.7	40.0	35.7
Lack of critical appraisal skills	21.9	23.1	0	28.6
Uncomfortable/unfamiliar with sources	9.4	0	0	14.3
Lack of trust/confidence in sources	9.4	7.7	0	7.1
Difficulty with language used in sources	9.4	0	0	21.4
Low technology literacy	6.3	0	0	7.1

(Continues)

TABLE 5 (Continued)

Barriers to drug information-seeking				
	Health care professionals ^b (N ^c = 32) % ^d	Physicians (N ^c = 13) % ^d	Nurses (N ^c = 5) % ^d	Pharmacists (N ^c = 14) % ^d
Fear of losing credibility to patients	3.1	0	20.0	0
Insufficient existing DI knowledge	3.1	0	0	7.1
<i>Environment characteristics</i>	43.8	46.2	20.0	42.9
Lack of time	37.5	46.2	0	35.7
Heavy workload	12.5	7.7	0	21.4
Lack of peer support	3.1	0	0	0
Working in isolation	3.1	0	20.0	0
Working at patient's home	3.1	0	20.0	0

^aExtracted data included in the analysis comprised of all data reported that is related to barriers to information-seeking, regardless of ranking.

^bHealth care professionals includes physicians, nurses and pharmacists.

^cNumber of articles, for each population, that reported on DI-seeking barriers that was included in this analysis.

^dPercentage of articles that reported on that category/subcategory of barrier. This is calculated using the frequency of articles reporting the respective categories/subcategories divided by the sum of articles included in the analysis for barriers for that population (refer to c).

- *MIMS* (Hall et al., 2003; Hassali et al., 2010; Lua et al., 2011; Ndosi & Newell, 2010; Rutter et al., 2004; Udezi et al., 2007; Wong et al., 2009) and
- *Drug Facts and Comparisons* (Carvajal et al., 2013; Moorman et al., 2017; Schrimsher et al., 2006).

Various drug databases were also mentioned, including:

- *MIMS* (Hall et al., 2003; Hassali et al., 2010; Ndosi & Newell, 2010; Rutter et al., 2004; Udezi et al., 2007; Wong et al., 2009),
- *UpToDate* (Lua et al., 2011; Moorman et al., 2017) and
- *Micromedex* (Carvajal et al., 2013; Moorman et al., 2017).

In addition, pharmaceutical companies/representatives (58.6%) were the most frequently mentioned human sources, followed by health care professionals (48.3%). Although primary sources ranked lower than tertiary and human sources, journals (58.6%) were the second most cited subcategory.

Tertiary sources and human sources were equally reported for physicians and nurses. However, tertiary sources (100%) were more frequently cited than human sources (61.5%) for pharmacists. Primary sources (journals) were also less frequently reported for pharmacists (30.8%), compared to physicians (80.0%) and nurses (75.0%). The top reported human source for physicians was 'Pharmaceutical companies/representatives' (80%), which only 50.0% and 38.5% of the studies on nurses and pharmacists respectively reported. In addition, Internet use by physicians (70.0%) and nurses (75.0%) was significantly higher than that of pharmacists (30.8%). All three

HCPs identified information services such as DI centre, but less than a quarter of the studies reported this.

Facilitators and barriers

Extracted data for facilitators and barriers were retrieved from 26 and 32 articles respectively. Five main themes and 21 subthemes were derived for facilitators while four main themes and 38 subthemes were derived for barriers. Results of the analyses are presented in Tables 4 and 5.

Facilitators

Facilitator themes most frequently reported for HCPs are *source* (92.3%), *access* (73.1%) and *individual characteristics* (34.6%). Meanwhile, the top cited subthemes are *credibility* (76.9%), *ease of use* (42.3%), *speed of use* (42.3%), *ease of accessing sources* (38.5%) and *adequate information in sources* (34.6%).

Upon comparison, it is found that *source characteristics* were the most highly reported facilitator theme across all three groups, followed by *access characteristics*. With regards to the subthemes, all three groups had 'credible source' as their top facilitator. It is also interesting to note that both physicians and pharmacists had 'ease of use' as one of their top three facilitators.

Physicians highly reported 'speed of use' (60.0%) as a facilitator but this was only reported by pharmacist and nurses in 25.0% and 33.3% of articles, respectively.

Meanwhile, 'sources are readily available' (41.7%) was one of pharmacists' top facilitator but was only cited 20.0% for physician and none for nurses.

Barriers

Source (84.4%), access (68.8%) and individual characteristics (65.6%) were also the top cited barrier themes for HCPs. The most prominent barrier subthemes include *issues with source credibility* (56.3%), *difficulties in accessing sources* (40.6%), *unavailability of sources* (37.5%), *insufficient knowledge regarding sources* (37.5%) and *lack of time* (37.5%).

A comparison of results of physicians and pharmacists found that *access* and *source characteristics* were highly reported (>50%) as barriers for both groups. However, *individual characteristics* (92.9%) were the most cited barriers for pharmacists, unlike for physicians which were least cited (38.5%).

On the other hand, *access*, *individual* and *source characteristics* were all equally cited barriers (40.0% each) for nurses.

Upon analysis, it was observed that there are two subthemes which are highly reported (within top 5) across all three groups (physicians, nurses and pharmacists)—*issues with source credibility* and *difficulties in accessing source*.

Other top reported subthemes for physicians are *lack of time* (46.1%) and *high or presence of cost* (38.5%). Meanwhile, the top cited subtheme for pharmacist is *insufficient knowledge regarding sources* (57.1%). Nurses also had other top reported subthemes like *unavailability of sources*, *insufficient knowledge regarding sources*, *lack of skills in searching* and *poor presentation of source* (each 40.0%).

DISCUSSION

This systematic literature review presents the findings on the drug ISB of physicians, nurses and pharmacists, as a whole and as separate groups, with the aim of understanding the similarities and differences of drug ISB amongst the different HCPs. This would facilitate the development of information services and education programs to meet the specific DI needs and resources of the HCPs. In doing so, this can potentially improve patient care and quality of health care services. A total of 48 articles reporting on the HCPs' DI needs, sources, facilitators and barriers were identified and analysed using content analysis.

Drug information needs

Results on DI needs revealed that HCPs have a wide range of DI requirements, in which our study identified

26 DI needs categories. However, it was observed that their top DI needs were similar, with 'Dosage/administration', 'Adverse drug reactions' and 'Drug interactions' being highly cited for all groups. This shows that, despite the wide range of DI needs, the different groups of HCPs have similar information needs—specifically, DI which are related to the prevention of drug therapy problems (Cipolle et al., 1998) such as adverse drug reaction and inappropriate dosage, to ensure the safe and efficacious use of drugs.

According to Leckie et al. (1996), information-seeking is a result of an 'information need' that is related to the individuals' work roles and tasks. This is evident from the differences in DI needs that our results identified.

For example, our review found that physicians but not pharmacists cited 'Efficacy', 'Cost and economics', and 'Comparisons with other drugs' as a DI need. A previous study (Ernst et al., 2000) similarly reported that physicians perceived information on cost as insufficient. This difference in information needs may be due to physicians' main task of prescribing (Lua et al., 2011). As per Sharifnia et al. (2018), prescribing decisions are influenced by multiple factors, which includes the cost and effectiveness of a drug.

In contrast, pharmacists but not physicians required DI on 'Pharmacology', 'Counselling' and 'Generic/Brand name'. This is unsurprising since pharmacists are considered the drug experts and are expected to be competent in the mechanism of actions of drugs. Moreover, since dispensing is one of pharmacists' main tasks (Ryan & Bond, 1994), information on counselling would be useful to provide a quick summary of main counselling points instead of having to screen through pages of information to pick out the relevant points. Pharmacists' role in medication reconciliation would also mean that they are required to know the generic/brand name in order to identify the drugs accurately.

In summary, it was found that the top DI needs were common across physicians, nurses and pharmacists. However, there were also DI needs which were relevant to a specific HCP group but not so to another, and this difference may be explained by their different work roles.

Drug information sources

Results on DI sources revealed that tertiary sources, such as reference books, the Internet and drug databases, were highly used by all three HCPs. This is unsurprising as the concise nature of tertiary sources allows HCPs to bypass their major information-seeking barriers like lack of time and lack of searching skills. These are barriers identified in this review (Table 5) and also reported by other

reviews studying physicians and nurses (Clarke et al., 2013; Kostagiolas et al., 2011; Younger, 2010). However, our review also found that users of tertiary sources may encounter issues such as reference books being outdated (Al-Farsi et al., 2014; Hall et al., 2003; Tumwikirize et al., 2008) or a lack of skills to assess quality of information on the Internet (Wong et al., 2009; Zehnder et al., 2004). Since tertiary sources are found to be most used, focusing on overcoming the limitations of these sources will be key to improving HCPs' DI seeking.

Human sources, such as pharmaceutical representatives and health care professionals, were also found to be the second most utilized DI sources by HCPs, which was similarly identified by Davies and Harrison (2007). The speed of use and low/no cost (Oshikoya et al., 2011) of these sources are likely reasons for its high usage. Dawes and Sampson (2003) shares that this finding may also be reflective of special needs of HCPs such as a psychological need for reassurance through the sharing of experiential knowledge from another individual.

It was also found that health care professionals tend to seek drug information from colleagues in the same profession. For example, the data showed that physicians prefer to seek drug information from their colleagues, seniors and consultants (Hedegaard & Damkier, 2009; Oshikoya et al., 2011; Sharma et al., 2017), while a small percentage of physicians approach the pharmacists for drug information requests (Lua et al., 2011). Likewise, pharmacists seem to gravitate towards their pharmacy colleagues when seeking drug information (Al-Farsi et al., 2014; Asmelashe Gelayee et al., 2017; Burgess et al., 2004; Wong et al., 2009). Approaching a fellow colleague is convenient and easily accessible, as opposed to finding another health care professional which one may not be familiar with. Nevertheless, some studies did report sources such as 'professional peers' or 'other health care professionals', which suggest that there is some interprofessional interaction when searching for drug information (Al-Farsi et al., 2014; Hassali et al., 2010). As for the nurses, there is currently insufficient evidence.

In addition, it was surprising that secondary resources were not mentioned since journals are accessed through bibliographic databases. A possible explanation may be that HCPs refer to searches on bibliographic databases as 'Internet' or were confused about the difference between search engines and databases (Davies & Harrison, 2007), potentially leading to misclassification. This lack of consistency in terminology was also raised by Younger (2010). Therefore, the use of secondary sources is likely under-represented and warrants a need for further investigation in future studies.

Pharmacists were found to use tertiary sources more than human sources, as compared to physicians and nurses, who cited both types of sources equally. A possible reason is that pharmacists are seen as the drug experts and hence, other health care professionals seek DI from the pharmacists but not the other way round. This is evident from the articles which cited pharmacists as a DI source of physicians (Abou-Auda, 2008; Hedegaard & Damkier, 2009; Lua et al., 2011; Murphy et al., 2006; Tumwikirize et al., 2008) and nurses (Hall et al., 2003; Murphy et al., 2006; Ndosi & Newell, 2010).

It is also noteworthy that physicians use pharmaceutical representatives significantly more than pharmacists and nurses. Since physicians' most prominent barrier is a lack of time (Table 5), the efficiency of pharmaceutical representatives may have resulted in their higher usage. Fickweiler et al. (2017) also found that physicians perceive pharmaceutical representatives as an important source of education. Physicians in a previous study however raised concerns about biased and misleading information from pharmaceutical representatives (Layton et al., 2007).

The lower use of the Internet by pharmacists was unexpected, given that Kostagiolas et al. (2011) cited the Internet as the top information source for pharmacists. This is likely because Kostagiolas et al. studied hospital pharmacists but half of the articles on pharmacists included in this review studied community pharmacists solely. The lack of Internet access by community pharmacists has been suggested by several studies (Al-Farsi et al., 2014; Burgess et al., 2004; McCaw et al., 2008; Usir et al., 2012).

Journals ranked second among the subcategories of DI sources, similar to the findings of Clarke et al.'s (2013) review on physicians and nurses. This may be due to the rise in the use of evidence-based medicine (EBM) (Hooker, 1997) which promotes the use of the best available research evidence. Usage of journals by pharmacists was however lower than that of physicians and nurses. Hennigen et al. (2009) postulated that the pharmacists' unfamiliarity and difficulty in access may have contributed to this low usage. However, this study was performed more than 10 years ago, and its results should be interpreted with caution, considering that trends may have changed.

Lastly, it was noted that information services such as DI centres were less mentioned by the HCPs. Potential reasons cited by various articles are the lack of awareness by HCPs regarding its availability (Abou-Auda, 2008), uncertainty of who can access these sources (Adibe, 2010) and the time-consuming process to obtain a response (Adibe, 2010; Goodwin et al., 2007; Hennigen et al., 2009).

All in all, the findings on the facilitators and barriers to drug ISB in this review was in agreement with the findings on DI sources used and provided further understanding on the results.

Facilitators and barriers to drug information-seeking

Results of facilitators and barriers reveal that *Source characteristics* were the most reported theme for the combined HCP group. Among the various source characteristics, *credibility* of the source was found to be the top facilitator and barrier subtheme for the combined HCP group. Across the physician, pharmacist and nurse groups, *credibility* of source was also the top facilitator subtheme and among the top five barrier subtheme. Credibility takes into consideration factors such as quality of information, reliability and trustworthiness of the source and, how updated and unbiased it is. The ever-changing nature of DI and the importance of accurate DI in providing quality health care make it unsurprising that HCPs rate source credibility as their top facilitator and barrier to DI-seeking.

Ease of use and *speed of use* were also facilitator subthemes that were frequently cited by HCPs. Moreover, both physicians and pharmacists had *ease of use* as one of their top three facilitators. This could be explained by the *lack of time* faced by HCPs, which is one of the top five cited barriers to DI-seeking. This lack of time was also similarly cited as a common barrier by other reviews on physicians' and nurses' information-seeking (Daei et al., 2020; Ebenezer, 2015). In a fast-paced working environment, it is no surprise that HCPs prioritize sources that are easy and quick to use, to obtain DI in a short period of time. However, it is also interesting to note that physicians cited *speed of use* as a facilitator more often than pharmacists and nurses, suggesting that physicians may face tighter time constraints than the other two HCP groups.

Access characteristics, such as the *availability* and *accessibility* of DI sources, is the second most reported theme in facilitators and in barriers for HCPs. This is possibly because the highly used tertiary sources such as reference books (e.g., British National Formulary) and drug databases (e.g., UpToDate) commonly require subscriptions or purchase costs. The accessibility of such sources is therefore dependent on the decision by institutions to purchase them.

Meanwhile, *ready availability of sources* was cited as a facilitator by pharmacists more than physicians and nurses. Of the five pharmacist articles which cited this, four are community pharmacist groups and one includes

both community and hospital pharmacists. This suggests that source availability is an important factor for community pharmacists and that there may be a lack of availability to DI sources by them, as supported by the community pharmacist articles (Asmelashe Gelayee et al., 2017; Burgess et al., 2004; Usir et al., 2012; Zehnder et al., 2004) which cited unavailability of sources as a barrier.

Also, *Individual characteristics*, which includes factors like *insufficient knowledge regarding sources* and a *lack of searching skills*, ranked first as barriers for pharmacists but last for physicians. These barriers were frequently mentioned in relation to the Internet (Burgess et al., 2004; Hennigen et al., 2009; Usir et al., 2012; Wong et al., 2009; Zehnder et al., 2004), possibly contributing to their low usage. A systematic review by Maclure and Stewart (2016) found that pharmacists want or need more digital literacy training, however recommendations from individual studies lack baseline data and are not current, quantifiable, measurable or specific. The study also highlighted the lack of evidence that digital literacy has been integrated into pharmacy staff training. This points towards a need for more studies on pharmacists' digital literacy knowledge levels and training.

In addition, it is surprising to note that a recent review by Daei et al. (2020) revealed different findings for the physicians—the lack of information-searching skills and unawareness of accessible sources were instead among the commonly cited barriers for physicians. This may suggest that over the recent years, searching skills and knowledge regarding accessible sources are increasingly an important factor for physicians to find information successfully.

It is also interesting to note that *insufficient knowledge regarding sources* and a *lack of searching skills* was also one of the top barrier subthemes for nurses. Ebenezer (2015) similarly found that a lack of confidence in searching was frequently cited for nurses. In a systematic review on nurses' barriers to using electronic evidence-based literature (Sadoughi et al., 2017), it was found that a lack of searching skills on bibliographic databases is the second most cited barrier. Evidence in the review also suggested that providing education on the use of electronic databases, search strategies and search features could help nurses search and retrieve drug information better from electronic databases. This point towards the importance of incorporating courses on search skills into the nursing curriculum.

Lastly, a notable finding is that the *high or presence of cost* was also a highly cited barrier for physicians. This is unsurprising since journals, the top cited DI source for physicians, commonly require high subscription or access costs.

RECOMMENDATIONS

Recommendations for educators

In view of the differences in DI needs, educators should tailor the content of their education programs according to their target HCPs' DI needs, which has been identified in this study. For example, educators in the medicine faculty may consider incorporating DI such as '*Efficacy*', '*Cost and economics*', '*Comparisons with other drugs*', '*Clinical trial results*', '*Use in diseased states*', '*Complementary & alternative medicine*', '*Drug overview*', '*Therapeutics*', '*Travel Medicine*' and '*General drug info*' in their curriculum. On the other hand, educators in the pharmacy faculty may find more value in emphasizing DI related to '*Pharmacology*', '*Counselling*' and '*Generic/Brand name*' in their curriculum.

The findings of this research can also guide educators in developing interprofessional education (IPE) programs. According to the World Health Organization (WHO, 2010), IPE occurs when two or more professions learn about, from and with each other to enable effective collaboration and improve health outcomes. Educators can tap on the similarities and differences of drug ISB between HCPs identified in this study to create IPE programs related to drug information. However, it is important to note that a curriculum in which the HCPs receive the same learning experience but without any reflective interaction among the students is not considered an IPE (Buring et al., 2009). Using the findings of this study and points raised in the Discussion section, the educators can seize the opportunity for students from different professions to discuss how their different work roles and tasks may shape their ISB related to drug information and how they can support one another in future workplace.

As suggested by the findings of this research, there is a potential need in the nursing and pharmacy school for more education on the available DI sources and the skills to use them. This includes digital literacy, which is especially important in the current digital era where resources are now found online or are available electronically. Moreover, educators should take into consideration the potential barriers faced in accessing these DI sources when the students graduate from school and no longer have the education institution's access to certain DI resources. Thus, it is important to introduce DI sources that have low or no cost or are readily accessible and available in the health care institutions.

In addition, DI centres were found to be not as well utilized. Usage of DI centres can bring about multiple benefits such as reducing drug-related problems and increased productivity (Umashankar et al., 2017).

Educators can play a role in promoting the use of DI centres by increasing awareness through their education programs.

Lastly, it was found that the most utilized DI sources are tertiary sources, possibly due to their ease and speed of use. However, some barriers related to these sources are that reference books may be outdated or there is a lack of skills to assess quality of information on the Internet. In view of the high usage of tertiary sources, it is important to educate HCPs on the risks and the precautions to take in retrieving information from such outdated sources. Educators should also teach HCPs how to assess the quality of information found on the Internet.

Recommendations for information providers

Similar to the recommendation for educators as mentioned above, information providers should also tailor the content of their services according to their target HCPs' DI needs, which has been identified in this study. Information providers should also utilize the similarities and differences in drug ISB between the HCP groups identified in this study to create effective interprofessional DI materials or references.

In addition, accessibility and availability of DI sources require further improvement. This can be achieved by reducing costs and improving distribution to institutions. Efforts should also be made to ensure HCPs are accessing the updated versions. Also, in places where community pharmacists lack Internet access, information providers may consider engaging them through human or paper-based sources which do not require Internet access.

Lastly, information providers should be aware of the lack of time faced by the HCPs. Improving the DI sources' ease of use and speed of use will facilitate HCPs' DI seeking and possibly encourage higher utilization of such sources. This could include friendly interfaces for easy navigation. Aside from that, information providers should ensure their sources are credible and provide quality, reliable, trustworthy, updated and unbiased information, which is a highly cited facilitator and barrier to DI seeking by the HCPs.

LIMITATIONS AND IMPLICATIONS FOR FUTURE RESEARCH

One methodological limitation is that articles unavailable in English have been excluded. Also, since most publications were from Asia, Europe and North America, the findings may be less applicable to other regions.

Moreover, the absence of critical appraisal may restrict readers' ability to judge the risk of bias of the included articles. This systematic literature review included studies from 1st January 2000 to 16th May 2020. A re-run of the search was performed in Jan 2022 and two new articles on drug ISB of physicians (Campbell et al., 2021; Jacquot et al., 2020) were found. While the data was not extracted and included in this systematic literature review, the findings are in line with the results of this review.

As there were limited studies for each country, with most only having one included study, sub-analysis by countries was not performed since results would be inconclusive given the small sample size.

The lack of consistency in design of the various studies—with varying sample sizes from 15 to 12229 and varying study types including quantitative, qualitative and mixed methods—also posed a challenge to the analysis of the results. This limitation was similarly raised by Younger (2010).

The number of included studies on nurses was also scarce. Nurses are the largest population of HCPs but only one-sixth of the included articles studied nurses. The drug ISB of HCPs hence represented that of physicians and pharmacists more than nurses. The lack of ISB studies on nurses was stressed by Younger (2010) almost a decade ago. It is alarming that the research on nurses' drug ISB is still scarce, and it is paramount that future research bridges this gap.

The findings of this research may also be used to guide recommendations for future research on the drug ISB of the HCPs. For example, since credibility was found to be highly cited as both a facilitator and barrier to drug ISB, future research may look into the types of DI sources which the HCPs find to be credible and the factors affecting their perception on the source credibility.

CONCLUSION

This study offers a comprehensive overview of the drug ISB of physicians, pharmacists and nurses. The DI needs, sources, facilitators and barriers of the three HCP groups were found to be similar. However, there were also differences that were evident. Information services and educators should take note of the similarities and differences in order to offer appropriate services and programs to the respective HCPs. The lack of ISB studies on nurses was also prominent and calls for greater attention to it.

AUTHOR CONTRIBUTIONS

Yu Xin Fiona Tan, Suzanne Tze Yin Lim and Jun Liang Lim screened and performed quality assessment of included studies. Yu Xin Fiona Tan and Suzanne Tze Yin Lim conducted the data extraction and analyses. All

authors were involved in the preparation, review and editing of the final manuscript.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX A

Table A1

TABLE A1 PRISMA 2020 checklist

Section and topic	Item #	Checklist item	Location where item is reported
<i>Title</i>			
Title	1	Identify the report as a systematic review.	P1—Title
<i>Abstract</i>			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1
<i>Introduction</i>			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P3 L1–5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P3 L19–21
<i>Methods</i>			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6 L1–10
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P5 L7, 15–17
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P6 L13–19
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P6 L22–25
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8–9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Table 1
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	N/A (P8 L8–10)

TABLE A1 (Continued)

Section and topic	Item #	Checklist item	Location where item is reported
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P6 L13–19
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P8 L2–8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P8–9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P8–9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A (P8 L8–10)
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A (P8 L8–10)
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P10, Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	N/A
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Tables 2–5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Tables 2–5

(Continues)

TABLE A1 (Continued)

Section and topic	Item #	Checklist item	Location where item is reported
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
<i>Discussion</i>			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P15–21
	23b	Discuss any limitations of the evidence included in the review.	P24 L6–12
	23c	Discuss any limitations of the review processes used.	P23 L22–25, P24 L1
	23d	Discuss implications of the results for practice, policy, and future research.	P21–23 P24 L13–23
<i>Other information</i>			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	N/A
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P25 L8–10
Competing interests	26	Declare any competing interests of review authors.	P25 L12–13
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	N/A

Note: From Page et al. (2021). For more information, visit: <http://www.prisma-statement.org/>.

APPENDIX B

Tables [B1–B12](#)

TABLE B1 Search strategy for PubMed (conducted 31 January 2019)

No.	Query	Results
#1	(Pharmacists[MeSH Terms]) OR Pharmacist*[Title/Abstract]	32,842
#2	(((((Physicians[MeSH Terms]) OR Physician*[Title/Abstract]) OR doctor*[Title/Abstract]) OR clinician*[Title/Abstract]) OR practitioner*[Title/Abstract])	796,473
#3	(Nurses[MeSH Terms]) OR Nurse*[Title/Abstract]	296,574
#4	((((((((((((((((((((((((((((((((((("Drug information"[Title/Abstract]) OR "Drugs information"[Title/Abstract]) OR "Medicine information"[Title/Abstract]) OR "Medicines information"[Title/Abstract]) OR "Medication information"[Title/Abstract]) OR "Medications information"[Title/Abstract]) OR "Medicinal information"[Title/Abstract]) OR "Drug informatic"[Title/Abstract]) OR "Drug informatics"[Title/Abstract]) OR "Prescribing information"[Title/Abstract]) OR "Drug dosing information"[Title/Abstract]) OR "Drug dosage information"[Title/Abstract]) OR "Drug dose information"[Title/Abstract]) OR "Drug interactions information"[Title/Abstract]) OR "Drug interaction information"[Title/Abstract]) OR "DDI information"[Title/Abstract]) OR "Drug administration information"[Title/Abstract]) OR "Adverse drug reaction information"[Title/Abstract]) OR "ADR information"[Title/Abstract]) OR "Over-the-counter information"[Title/Abstract]) OR "Drug indication information"[Title/Abstract]) OR "Drug content information"[Title/Abstract]) OR "Pharmacokinetic information"[Title/Abstract]) OR "Pharmacokinetics information"[Title/Abstract]) OR "Drug compatibility information"[Title/Abstract]) OR "Drug stability information"[Title/Abstract]) OR "Drug contraindication information"[Title/Abstract]) OR "Drug contraindications information"[Title/Abstract]) OR "Pharmacology information"[Title/Abstract]) OR "MOA information"[Title/Abstract]) OR "Mechanism of action information"[Title/Abstract]) OR "Drug pregnancy information"[Title/Abstract]) OR "Drug lactation information"[Title/Abstract]) OR "Drug identification information"[Title/Abstract]) OR "Drug cost information"[Title/Abstract]) OR "Drug price information"[Title/Abstract]) OR "Drug therapy information"[Title/Abstract]) OR "Dispensing information"[Title/Abstract]) OR "Therapy information"[Title/Abstract]) OR "Pharmaceutic information"[Title/Abstract]) OR "Pharmaceutics information"[Title/Abstract]) OR "Pharmaceutical information"[Title/Abstract]) OR "Pharmacologic information"[Title/Abstract]) OR "Pharmacologics information"[Title/Abstract]) OR "Pharmacological information"[Title/Abstract]	5626
#5	((((((((((((((((information seeking behavior[MeSH Terms]) OR information literacy[MeSH Terms]) OR Information seeking behavio*[Title/Abstract]) OR Information seek*[Title/Abstract]) OR Information sourc*[Title/Abstract]) OR Information search*[Title/Abstract]) OR Information forag*[Title/Abstract]) OR Information retriev*[Title/Abstract]) OR Information gather*[Title/Abstract]) OR Information behavio*[Title/Abstract]) OR Information acqui*[Title/Abstract]) OR Information expect*[Title/Abstract]) OR "Seeking information"[Title/Abstract]) OR "Seek information"[Title/Abstract]) OR "Seek for information"[Title/Abstract]) OR "Seeking for information"[Title/Abstract]) OR "Gather information"[Title/Abstract]) OR "Seeking knowledge"[Title/Abstract]) OR "Seek knowledge"[Title/Abstract]) OR "Acquire knowledge"[Title/Abstract])	19,929
#6	((((((((((needs assessment[MeSH Terms]) OR Information need*[Title/Abstract]) OR Information require*[Title/Abstract]) OR Information demand*[Title/Abstract]) OR Information concern*[Title/Abstract]) OR Information preference*[Title/Abstract]) OR information gap*[Title/Abstract]) OR Information wish*[Title/Abstract]) OR Information ask*[Title/Abstract]) OR Prescribing need*[Title/Abstract]) OR Knowledge need*[Title/Abstract]) OR Informational need*[Title/Abstract])	78,452
#7	((((((((((((((drug information services[MeSH Terms]) OR information services[MeSH Terms]) OR Information source*[Title/Abstract]) OR Information resource*[Title/Abstract]) OR Information reference*[Title/Abstract]) OR Information material*[Title/Abstract]) OR Information service*[Title/Abstract]) OR "Source of information"[Title/Abstract]) OR "Sources of information"[Title/Abstract]) OR "Source of drug information"[Title/Abstract]) OR "Sources of drug information"[Title/Abstract]) OR Drug reference*[Title/Abstract]) OR Drug resource*[Title/Abstract]) OR Drug database*[Title/Abstract])	979,159
#8	(((((Information barrier*[Title/Abstract]) OR Information obstacle*[Title/Abstract]) OR Information challeng*[Title/Abstract]) OR Information facilitat*[Title/Abstract]) OR Information enabl*[Title/Abstract])	262
#9	#1 OR #2 OR #3	1,055,895
#10	#5 OR #6 OR #7 OR #8	1,065,234
#11	#4 AND #9 AND #10	1085
#12	#4 AND #9 AND #10	576

Filters: Publication date from 2000/01/01 to 2019/12/31; English

TABLE B2 Search strategy for PubMed (conducted 16 May 2020)

No.	Query	Results
#1	(Pharmacists[MeSH Terms]) OR Pharmacist*[Title/Abstract]	36,658
#2	(((((Physicians[MeSH Terms]) OR Physician*[Title/Abstract]) OR doctor*[Title/Abstract]) OR clinician*[Title/Abstract]) OR practitioner*[Title/Abstract])	865,235
#3	(Nurses[MeSH Terms]) OR Nurse*[Title/Abstract]	314,540
#4	((((((((((((((((((((((((((((((("Drug information"[Title/Abstract]) OR "Drugs information"[Title/Abstract]) OR "Medicine information"[Title/Abstract]) OR "Medicines information"[Title/Abstract]) OR "Medication information"[Title/Abstract]) OR "Medications information"[Title/Abstract]) OR "Medicinal information"[Title/Abstract]) OR "Drug informatic"[Title/Abstract]) OR "Drug informatics"[Title/Abstract]) OR "Prescribing information"[Title/Abstract]) OR "Drug dosing information"[Title/Abstract]) OR "Drug dosage information"[Title/Abstract]) OR "Drug dose information"[Title/Abstract]) OR "Drug interactions information"[Title/Abstract]) OR "Drug interaction information"[Title/Abstract]) OR "DDI information"[Title/Abstract]) OR "Drug administration information"[Title/Abstract]) OR "Adverse drug reaction information"[Title/Abstract]) OR "ADR information"[Title/Abstract]) OR "Over-the-counter information"[Title/Abstract]) OR "Drug indication information"[Title/Abstract]) OR "Drug content information"[Title/Abstract]) OR "Pharmacokinetic information"[Title/Abstract]) OR "Pharmacokinetics information"[Title/Abstract]) OR "Drug compatibility information"[Title/Abstract]) OR "Drug stability information"[Title/Abstract]) OR "Drug contraindication information"[Title/Abstract]) OR "Drug contraindications information"[Title/Abstract]) OR "Pharmacology information"[Title/Abstract]) OR "MOA information"[Title/Abstract]) OR "Mechanism of action information"[Title/Abstract]) OR "Drug pregnancy information"[Title/Abstract]) OR "Drug lactation information"[Title/Abstract]) OR "Drug identification information"[Title/Abstract]) OR "Drug cost information"[Title/Abstract]) OR "Drug price information"[Title/Abstract]) OR "Drug therapy information"[Title/Abstract]) OR "Dispensing information"[Title/Abstract]) OR "Therapy information"[Title/Abstract]) OR "Pharmaceutic information"[Title/Abstract]) OR "Pharmaceutics information"[Title/Abstract]) OR "Pharmaceutical information"[Title/Abstract]) OR "Pharmacologic information"[Title/Abstract]) OR "Pharmacological information"[Title/Abstract])	6019
#5	((((((((((((((((((((((((((((((information seeking behavior[MeSH Terms]) OR information literacy[MeSH Terms]) OR Information seeking behavio*[Title/Abstract]) OR Information seek*[Title/Abstract]) OR Information sourc*[Title/Abstract]) OR Information search*[Title/Abstract]) OR Information forag*[Title/Abstract]) OR Information retriev*[Title/Abstract]) OR Information gather*[Title/Abstract]) OR Information behavio*[Title/Abstract]) OR Information acqui*[Title/Abstract]) OR Information expect*[Title/Abstract]) OR "Seeking information"[Title/Abstract]) OR "Seek information"[Title/Abstract]) OR "Seek for information"[Title/Abstract]) OR "Seeking for information"[Title/Abstract]) OR "Gather information"[Title/Abstract]) OR "Seeking knowledge"[Title/Abstract]) OR "Seek knowledge"[Title/Abstract]) OR "Acquire knowledge"[Title/Abstract])	22,885
#6	((((((((((needs assessment[MeSH Terms]) OR Information need*[Title/Abstract]) OR Information require*[Title/Abstract]) OR Information demand*[Title/Abstract]) OR Information concern*[Title/Abstract]) OR Information preference*[Title/Abstract]) OR information gap*[Title/Abstract]) OR Information wish*[Title/Abstract]) OR Information ask*[Title/Abstract]) OR Prescribing need*[Title/Abstract]) OR Knowledge need*[Title/Abstract]) OR Informational need*[Title/Abstract])	84,369
#7	((((((((((((((drug information services[MeSH Terms]) OR information services[MeSH Terms]) OR Information source*[Title/Abstract]) OR Information resource*[Title/Abstract]) OR Information reference*[Title/Abstract]) OR Information material*[Title/Abstract]) OR Information service*[Title/Abstract]) OR "Source of information"[Title/Abstract]) OR "Sources of information"[Title/Abstract]) OR "Source of drug information"[Title/Abstract]) OR "Sources of drug information"[Title/Abstract]) OR Drug reference*[Title/Abstract]) OR Drug resource*[Title/Abstract]) OR Drug database*[Title/Abstract])	1,001,672
#8	(((((Information barrier*[Title/Abstract]) OR Information obstacle*[Title/Abstract]) OR Information challeng*[Title/Abstract]) OR Information facilitat*[Title/Abstract]) OR Information enabl*[Title/Abstract])	275
#9	#1 OR #2 OR #3	1,140,238
#10	#5 OR #6 OR #7 OR #8	1,095,567
#11	#4 AND #9 AND #10	1127
#12	#4 AND #9 AND #10 Filters: Publication date from 2019/02/01 to 2020/05/16; English	39

TABLE B3 Search strategy for Embase (conducted 31 January 2019)

No.	Query	Results
#1	'pharmacist'/exp OR 'pharmacist*':ab,ti	92,063
#2	'physician'/exp OR 'clinician'/exp OR 'health practitioner'/exp OR 'physician*':ab,ti OR 'doctor*':ab,ti OR 'clinician*':ab,ti OR 'practitioner*':ab,ti	1,418,763
#3	'nurse'/exp OR 'nurse*':ab,ti	358,463
#4	'drug information'/exp OR 'drug information' OR 'drug information':ab,ti OR 'drugs information':ab,ti OR 'medicine information':ab,ti OR 'medicines information':ab,ti OR 'medication information':ab,ti OR 'medications information':ab,ti OR 'medicinal information':ab,ti OR 'drug informatic':ab,ti OR 'drug informatics':ab,ti OR 'prescribing information':ab,ti OR 'drug dosing information':ab,ti OR 'drug dosage information':ab,ti OR 'drug dose information':ab,ti OR 'drug interactions information':ab,ti OR 'drug interaction information':ab,ti OR 'ddi information':ab,ti OR 'drug administration information':ab,ti OR 'adverse drug reaction information':ab,ti OR 'adr information':ab,ti OR 'over-the-counter information':ab,ti OR 'drug indication information':ab,ti OR 'drug content information':ab,ti OR 'pharmacokinetic information':ab,ti OR 'pharmacokinetics information':ab,ti OR 'drug compatibility information':ab,ti OR 'drug stability information':ab,ti OR 'drug contraindication information':ab,ti OR 'drug contraindications information':ab,ti OR 'pharmacology information':ab,ti OR 'moa information':ab,ti OR 'mechanism of action information':ab,ti OR 'drug pregnancy information':ab,ti OR 'drug lactation information':ab,ti OR 'drug identification information':ab,ti OR 'drug cost information':ab,ti OR 'drug price information':ab,ti OR 'drug therapy information':ab,ti OR 'dispensing information':ab,ti OR 'therapy information':ab,ti OR 'pharmaceutic information':ab,ti OR 'pharmaceutics information':ab,ti OR 'pharmaceutical information':ab,ti OR 'pharmacologic information':ab,ti OR 'pharmacologics information':ab,ti OR 'pharmacological information':ab,ti	29,458
#5	'information seeking'/exp OR 'information literacy'/exp OR 'information seeking behavio*':ab,ti OR 'information seek*':ab,ti OR 'information sourc*':ab,ti OR 'information search*':ab,ti OR 'information forag*':ab,ti OR 'information retriev*':ab,ti OR 'information gather*':ab,ti OR 'information behavio*':ab,ti OR 'information acqui*':ab,ti OR 'information expect*':ab,ti OR 'seeking information':ab,ti OR 'seek information':ab,ti OR 'seek for information':ab,ti OR 'seeking for information':ab,ti OR 'gather information':ab,ti OR 'seeking knowledge':ab,ti OR 'seek knowledge':ab,ti OR 'acquire knowledge':ab,ti	24,313
#6	'needs assessment'/exp OR 'information need*':ab,ti OR 'information require*':ab,ti OR 'information demand*':ab,ti OR 'information concern*':ab,ti OR 'information preference*':ab,ti OR 'information gap*':ab,ti OR 'information wish*':ab,ti OR 'information ask*':ab,ti OR 'prescribing need*':ab,ti OR 'knowledge need*':ab,ti OR 'informational need*':ab,ti	50,684
#7	'drug database'/exp OR 'information service'/exp OR 'information source*':ab,ti OR 'information resource*':ab,ti OR 'information reference*':ab,ti OR 'information material*':ab,ti OR 'information service*':ab,ti OR 'source of information':ab,ti OR 'sources of information':ab,ti OR 'source of drug information':ab,ti OR 'sources of drug information':ab,ti OR 'drug reference*':ab,ti OR 'drug resource*':ab,ti OR 'drug database*':ab,ti	43,517
#8	'information barrier*':ab,ti OR 'information obstacle*':ab,ti OR 'information challeng*':ab,ti OR 'information facilitat*':ab,ti OR 'information enabl*':ab,ti	1097
#9	#1 OR #2 OR #3	1,743,057
#10	#5 OR #6 OR #7 OR #8	110,307
#11	#4 AND #9 AND #10	1503
#12	#4 AND #9 AND #10 AND [english]/lim AND [2000-2019]/py	947

TABLE B4 Search strategy for Embase (conducted 16 May 2020)

No.	Query	Results
#1	'pharmacist'/exp OR 'pharmacist*':ab,ti	99,854
#2	'physician'/exp OR 'clinician'/exp OR 'health practitioner'/exp OR 'physician*':ab,ti OR 'doctor*':ab,ti OR 'clinician*':ab,ti OR 'practitioner*':ab,ti	1,570,264
#3	'nurse'/exp OR 'nurse*':ab,ti	386,313
#4	'drug information'/exp OR 'drug information' OR 'drug information':ab,ti OR 'drugs information':ab,ti OR 'medicine information':ab,ti OR 'medicines information':ab,ti OR 'medication information':ab,ti OR 'medications information':ab,ti OR 'medicinal information':ab,ti OR 'drug informatic':ab,ti OR 'drug informatics':ab,ti OR 'prescribing information':ab,ti OR 'drug dosing information':ab,ti OR 'drug dosage information':ab,ti OR 'drug dose information':ab,ti OR 'drug interactions information':ab,ti OR 'drug interaction information':ab,ti OR 'ddi information':ab,ti OR 'drug administration information':ab,ti OR 'adverse drug reaction information':ab,ti OR 'adr information':ab,ti OR 'over-the-counter information':ab,ti OR 'drug indication information':ab,ti OR 'drug content information':ab,ti OR 'pharmacokinetic information':ab,ti OR 'pharmacokinetics information':ab,ti OR 'drug compatibility information':ab,ti OR 'drug stability information':ab,ti OR 'drug contraindication information':ab,ti OR 'drug contraindications information':ab,ti OR 'pharmacology information':ab,ti OR 'moa information':ab,ti OR 'mechanism of action information':ab,ti OR 'drug pregnancy information':ab,ti OR 'drug lactation information':ab,ti OR 'drug identification information':ab,ti OR 'drug cost information':ab,ti OR 'drug price information':ab,ti OR 'drug therapy information':ab,ti OR 'dispensing information':ab,ti OR 'therapy information':ab,ti OR 'pharmaceutic information':ab,ti OR 'pharmaceutics information':ab,ti OR 'pharmaceutical information':ab,ti OR 'pharmacologic information':ab,ti OR 'pharmacologics information':ab,ti OR 'pharmacological information':ab,ti	37,284
#5	'information seeking'/exp OR 'information literacy'/exp OR 'information seeking behavio*':ab,ti OR 'information seek*':ab,ti OR 'information sourc*':ab,ti OR 'information search*':ab,ti OR 'information forag*':ab,ti OR 'information retriev*':ab,ti OR 'information gather*':ab,ti OR 'information behavio*':ab,ti OR 'information acqui*':ab,ti OR 'information expect*':ab,ti OR 'seeking information':ab,ti OR 'seek information':ab,ti OR 'seek for information':ab,ti OR 'seeking for information':ab,ti OR 'gather information':ab,ti OR 'seeking knowledge':ab,ti OR 'seek knowledge':ab,ti OR 'acquire knowledge':ab,ti	27,155
#6	'needs assessment'/exp OR 'information need*':ab,ti OR 'information require*':ab,ti OR 'information demand*':ab,ti OR 'information concern*':ab,ti OR 'information preference*':ab,ti OR 'information gap*':ab,ti OR 'information wish*':ab,ti OR 'information ask*':ab,ti OR 'prescribing need*':ab,ti OR 'knowledge need*':ab,ti OR 'informational need*':ab,ti	54,360
#7	'drug database'/exp OR 'information service'/exp OR 'information source*':ab,ti OR 'information resource*':ab,ti OR 'information reference*':ab,ti OR 'information material*':ab,ti OR 'information service*':ab,ti OR 'source of information':ab,ti OR 'sources of information':ab,ti OR 'source of drug information':ab,ti OR 'sources of drug information':ab,ti OR 'drug reference*':ab,ti OR 'drug resource*':ab,ti OR 'drug database*':ab,ti	46,818
#8	'information barrier*':ab,ti OR 'information obstacle*':ab,ti OR 'information challeng*':ab,ti OR 'information facilitat*':ab,ti OR 'information enabl*':ab,ti	1236
#9	#1 OR #2 OR #3	1,916,568
#10	#5 OR #6 OR #7 OR #8	119,340
#11	#4 AND #9 AND #10	1591
#12	#4 AND #9 AND #10 AND [english]/lim AND [2019-2020]/py	68

TABLE B5 Search strategy for SCOPUS (conducted 31 January 2019)

No.	Query	Results
#1	TITLE-ABS-KEY (pharmacist* OR physician* OR doctor* OR clinician* OR practitioner* OR nurse*)	1,741,394
#2	TITLE-ABS-KEY ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information")	31,444
#3	TITLE-ABS-KEY ("Information seeking behavio*" OR "Information seek*" OR "Information sourc*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavio*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge" OR "Information Literacy")	217,713
#4	TITLE-ABS-KEY ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*" OR "Needs Assessment")	109,031
#5	TITLE-ABS-KEY ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*" OR "Information Service*")	147,256
#6	TITLE-ABS-KEY ("Information barrier*" OR "Information obstacle*" OR "Information challeng*" OR "Information facilitat*" OR "Information enabl*")	2984
#7	#3 OR #4 OR #5 OR #6	433,458
#8	#1 AND #2 AND #7	2583
#9	#8 AND PUBYEAR > 1999	1532
#10	#8 AND PUBYEAR > 1999 AND (LIMIT-TO (LANGUAGE, "English"))	1283

TABLE B6 Search strategy for SCOPUS (conducted 16 May 2020)

No.	Query	Results
#1	TITLE-ABS-KEY (pharmacist* OR physician* OR doctor* OR clinician* OR practitioner* OR nurse*)	1,878,110
#2	TITLE-ABS-KEY ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information")	32,453
#3	TITLE-ABS-KEY ("Information seeking behavior*" OR "Information seek*" OR "Information source*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavior*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge" OR "Information Literacy")	235,661
#4	TITLE-ABS-KEY ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*" OR "Needs Assessment")	117,519
#5	TITLE-ABS-KEY ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*" OR "Information Service*")	158,688
#6	TITLE-ABS-KEY ("Information barrier*" OR "Information obstacle*" OR "Information challeng*" OR "Information facilitat*" OR "Information enabl*")	3353
#7	#3 OR #4 OR #5 OR #6	468,669
#8	#1 AND #2 AND #7	2622
#9	#8 AND PUBYEAR > 2018 AND (LIMIT-TO (LANGUAGE, "English"))	64

TABLE B7 Search strategy for PsycINFO (via Ovid) (conducted 31 January 2019)

No.	Query	Results
#1	exp Pharmacists/ or (Pharmacist*).ti,ab.	2983
#2	exp Physicians/ or exp Clinicians/ or (physician* or doctor* or clinician* or practitioner*).ti,ab.	257,454
#3	exp Nurses/ or (Nurse*).ti,ab.	64,832
#4	(Drug information or Drugs information or Medicine information or Medicines information or Medication information or Medications information or Medicinal information or Drug informatic or Drug informatics or Prescribing information or Drug dosing information or Drug dosage information or Drug dose information or Drug interactions information or Drug interaction information or DDI information or Drug administration information or Adverse drug reaction information or ADR information or Over-the-counter information or Drug indication information or Drug content information or Pharmacokinetic information or Pharmacokinetics information or Drug compatibility information or Drug stability information or Drug contraindication information or Drug contraindications information or Pharmacology information or MOA information or Mechanism of action information or Drug pregnancy information or Drug Lactation information or Drug identification information or Drug cost information or Drug price information or Drug therapy information or Dispensing information or Therapy information or Pharmaceutic information or Pharmaceuticals information or Pharmaceutical information or Pharmacologic information or Pharmacologics information or Pharmacological information).ti,ab.	803
#5	exp Information Seeking/ or exp Information Literacy/ or (Information seeking behavio* or Information seek* or Information sourc* or Information search* or Information forag* or Information retriev* or Information gather* or Information behavio* or Information acqui* or Information expect* or Seeking information or Seek information or Seek for information or Seeking for information or Gather information or Seeking knowledge or Seek knowledge or Acquire knowledge).ti,ab.	17,064
#6	exp Needs Assessment/ or (Information need* or Information require* or Information demand* or Information concern* or Information preference* or Information gap* or Information wish* or Information ask* or Prescribing need* or Knowledge need* or Informational need*).ti,ab.	12,433
#7	exp Information Services/ or (Information source* or Information resource* or Information reference* or Information material* or Information service* or Source of information or Sources of information or Source of drug information or Sources of drug information or Drug reference* or Drug resource* or Drug database*).ti,ab.	12,303
#8	(Information barrier* or Information obstacle* or Information challeng* or Information facilitat* or Information enabl*).ti,ab.	413
#9	#1 OR #2 OR #3	308,736
#10	#5 OR #6 OR #7 OR #8	38,049
#11	#4 AND #9 AND #10	91
#12	limit 11 to (english language and yr="2000 -2019")	77

TABLE B8 Search strategy for PsycINFO (via Ovid) (conducted 16 May 2020)

No.	Query	Results
#1	exp Pharmacists/ or (Pharmacist*).ti,ab.	3231
#2	exp Physicians/ or exp Clinicians/ or (physician* or doctor* or clinician* or practitioner*).ti,ab.	271,694
#3	exp Nurses/ or (Nurse*).ti,ab.	68,599
#4	(Drug information or Drugs information or Medicine information or Medicines information or Medication information or Medications information or Medicinal information or Drug informatic or Drug informatics or Prescribing information or Drug dosing information or Drug dosage information or Drug dose information or Drug interactions information or Drug interaction information or DDI information or Drug administration information or Adverse drug reaction information or ADR information or Over-the-counter information or Drug indication information or Drug content information or Pharmacokinetic information or Pharmacokinetics information or Drug compatibility information or Drug stability information or Drug contraindication information or Drug contraindications information or Pharmacology information or MOA information or Mechanism of action information or Drug pregnancy information or Drug Lactation information or Drug identification information or Drug cost information or Drug price information or Drug therapy information or Dispensing information or Therapy information or Pharmaceutic information or Pharmaceuticals information or Pharmaceutical information or Pharmacologic information or Pharmacologics information or Pharmacological information).ti,ab.	840
#5	exp Information Seeking/ or exp Information Literacy/ or (Information seeking behavio* or Information seek* or Information sourc* or Information search* or Information forag* or Information retriev* or Information gather* or Information behavio* or Information acqui* or Information expect* or Seeking information or Seek information or Seek for information or Seeking for information or Gather information or Seeking knowledge or Seek knowledge or Acquire knowledge).ti,ab.	20,922
#6	exp Needs Assessment/ or (Information need* or Information require* or Information demand* or Information concern* or Information preference* or Information gap* or Information wish* or Information ask* or Prescribing need* or Knowledge need* or Informational need*).ti,ab.	12,999
#7	exp Information Services/ or (Information source* or Information resource* or Information reference* or Information material* or Information service* or Source of information or Sources of information or Source of drug information or Sources of drug information or Drug reference* or Drug resource* or Drug database*).ti,ab.	12,958
#8	(Information barrier* or Information obstacle* or Information challeng* or Information facilitat* or Information enabl*).ti,ab.	428
#9	#1 OR #2 OR #3	326,009
#10	#5 OR #6 OR #7 OR #8	42,855
#11	#4 AND #9 AND #10	91
#12	limit 11 to (english language and yr="2019 -Current")	0

TABLE B9 Search strategy for CINAHL Plus with Full Text (via EBSCOhost) (conducted 31 January 2019)

No.	Query	Results
#1	(MH ("Pharmacists")) OR (TI ("pharmacist*")) OR (AB ("pharmacist*"))	19,918
#2	(MH ("Physicians+")) OR (TI ("physician*" OR "doctor*" OR "clinician*" OR "practitioner*")) OR (AB ("physician*" OR "doctor*" OR "clinician*" OR "practitioner*"))	347,148
#3	(MH ("Nurses+")) OR (TI ("nurse*")) OR (AB ("nurse*"))	403,096
#4	(MH ("Drug Information")) OR (TI ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologies information" OR "Pharmacological information")) OR (AB ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologies information" OR "Pharmacological information"))	3473
#5	(MH ("Information Seeking Behavior" OR "Information Literacy+")) OR (TI ("Information seeking behavio*" OR "Information seek*" OR "Information sourc*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavio*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge")) OR (AB ("Information seeking behavio*" OR "Information seek*" OR "Information sourc*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavio*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge"))	16,488
#6	(MH ("Information Needs" OR "Needs Assessment")) OR (TI ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*")) OR (AB ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*"))	28,917
#7	(MH ("Information Services+" OR "Drug Information Services+" OR "Information Resources+" OR "Reference Databases+" OR "Resource Databases+")) OR (TI ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Information service*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*")) OR (AB ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Information service*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*"))	465,217

(Continues)

TABLE B9 (Continued)

No.	Query	Results
	"Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*")	
#8	(TI ("Information barrier*" OR "Information obstacle*" OR "Information challeng*" OR "Information facilitat*" OR "Information enabl*")) OR (AB ("Information barrier*" OR "Information obstacle*" OR "Information challeng*" OR "Information facilitat*" OR "Information enabl*"))	263
#9	#1 OR #2 OR #3	714,514
#10	#5 OR #6 OR #7 OR #8	495,776
#11	#4 AND #9 AND #10	513
#12	#11 (Limiters - Published Date: 20000101-20191231) (Narrow by Language: - english)	443

TABLE B10 Search strategy for CINAHL Plus with Full Text (via EBSCOhost) (conducted 16 May 2020)

No.	Query	Results
#1	(MH ("Pharmacists")) OR (TI ("pharmacist*")) OR (AB ("pharmacist*"))	26,598
#2	(MH ("Physicians+")) OR (TI ("physician*" OR "doctor*" OR "clinician*" OR "practitioner*")) OR (AB ("physician*" OR "doctor*" OR "clinician*" OR "practitioner*"))	456,183
#3	(MH ("Nurses+")) OR (TI ("nurse*")) OR (AB ("nurse*"))	494,746
#4	(MH ("Drug Information")) OR (TI ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information")) OR (AB ("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information"))	4512
#5	(MH ("Information Seeking Behavior" OR "Information Literacy+")) OR (TI ("Information seeking behavio*" OR "Information seek*" OR "Information sourc*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavio*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge")) OR (AB ("Information seeking behavio*" OR "Information seek*" OR "Information sourc*" OR "Information search*" OR "Information forag*" OR "Information retriev*" OR "Information gather*" OR "Information behavio*" OR "Information acqui*" OR "Information expect*" OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge"))	21,705
#6	(MH ("Information Needs" OR "Needs Assessment")) OR (TI ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*")) OR (AB ("Information need*" OR "Information require*" OR "Information demand*" OR "Information concern*" OR "Information preference*" OR "Information gap*" OR "Information wish*" OR "Information ask*" OR "Prescribing need*" OR "Knowledge need*" OR "Informational need*"))	36,154
#7	(MH ("Information Services+" OR "Drug Information Services+" OR "Information Resources+" OR "Reference Databases+" OR "Resource Databases+")) OR (TI ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Information service*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*")) OR (AB ("Information source*" OR "Information resource*" OR "Information reference*" OR "Information material*" OR "Information service*" OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR "Drug reference*" OR "Drug resource*" OR "Drug database*"))	586,326

(Continues)

TABLE B10 (Continued)

No.	Query	Results
	“Sources of information” OR “Source of drug information” OR “Sources of drug information” OR “Drug reference*” OR “Drug resource*” OR “Drug database*”)	
#8	(TI (“Information barrier*” OR “Information obstacle*” OR “Information challeng*” OR “Information facilitat*” OR “Information enabl*”)) OR (AB (“Information barrier*” OR “Information obstacle*” OR “Information challeng*” OR “Information facilitat*” OR “Information enabl*”))	332
#9	#1 OR #2 OR #3	905,168
#10	#5 OR #6 OR #7 OR #8	625,245
#11	#4 AND #9 AND #10	690
#12	#11 (Limiters - Published Date: 20190101-20201231) (Narrow by Language: - english)	59

TABLE B11 Search strategy for COCHRANE LIBRARY (via Wiley) (conducted 31 January 2019)

No.	Query	Results
#1	MeSH descriptor: [Pharmacists] explode all trees	517
#2	(Pharmacist*):ti,ab	3076
#3	MeSH descriptor: [Physicians] explode all trees	1805
#4	(Physician* OR Doctor* OR Clinician* OR Practitioner*):ti,ab	53,436
#5	MeSH descriptor: [Nurses] explode all trees	1094
#6	(Nurse*):ti,ab	16,663
#7	#1 OR #2 OR #3 OR #4 OR #5 OR #6	67,355
#8	("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information"):ti,ab	506
#9	MeSH descriptor: [Information Seeking Behavior] explode all trees	39
#10	MeSH descriptor: [Information Literacy] explode all trees	294
#11	(Information NEXT seeking NEXT behavio* OR Information NEXT seek* OR Information NEXT sourc* OR Information NEXT search* OR Information NEXT forag* OR Information NEXT retriev* OR Information NEXT gather* OR Information NEXT behavio* OR Information NEXT acqui* OR Information NEXT expect* OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge"):ti,ab	803
#12	#9 OR #10 OR #11	1112
#13	MeSH descriptor: [Needs Assessment] explode all trees	337
#14	(Information NEXT need* OR Information NEXT require* OR Information NEXT demand* OR Information NEXT concern* OR Information NEXT preference* OR information NEXT gap* OR Information NEXT wish* OR Information NEXT ask* OR Prescribing NEXT need* OR Knowledge NEXT need* OR Informational NEXT need*):ti,ab	834
#15	#13 OR #14	1167
#16	MeSH descriptor: [Information Services] explode all trees	1891
#17	MeSH descriptor: [Drug Information Services] explode all trees	147
#18	(Information NEXT source* OR Information NEXT resource* OR Information NEXT reference* OR Information NEXT material* OR Information NEXT service* OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR Drug NEXT reference* OR Drug NEXT resource* OR Drug NEXT database*):ti,ab	837
#19	#16 OR #17 OR #18	2835
#20	(Information NEXT barrier* OR Information NEXT obstacle* OR Information NEXT challeng* OR Information NEXT facilitat* OR Information NEXT enabl*):ti,ab	53
#21	#12 OR #15 OR #19 OR #20	4966
#22	#7 AND #8 AND #21	27
#23	#7 AND #8 AND #21 (with COCHRANE LIBRARY publication date from Jan 2000 to Dec 2019)	25

TABLE B12 Search strategy for COCHRANE LIBRARY (via Wiley) (conducted 16 May 2020)

No.	Query	Results
#1	MeSH descriptor: [Pharmacists] explode all trees	563
#2	(Pharmacist*):ti,ab	4001
#3	MeSH descriptor: [Physicians] explode all trees	1950
#4	(Physician* OR Doctor* OR Clinician* OR Practitioner*):ti,ab	74,035
#5	MeSH descriptor: [Nurses] explode all trees	1167
#6	(Nurse*):ti,ab	23,243
#7	#1 OR #2 OR #3 OR #4 OR #5 OR #6	92,872
#8	("Drug Information" OR "Drugs Information" OR "Medicine Information" OR "Medicines Information" OR "Medication Information" OR "Medications Information" OR "Medicinal Information" OR "Drug Informatic" OR "Drug Informatics" OR "Prescribing Information" OR "Drug dosing Information" OR "Drug dosage information" OR "Drug dose information" OR "Drug Interactions Information" OR "Drug Interaction Information" OR "DDI Information" OR "Drug administration Information" OR "Adverse drug reaction information" OR "ADR Information" OR "Over-the-counter Information" OR "Drug indication information" OR "Drug Content Information" OR "Pharmacokinetic Information" OR "Pharmacokinetics Information" OR "Drug compatibility information" OR "Drug stability information" OR "Drug contraindication information" OR "Drug contraindications information" OR "Pharmacology information" OR "MOA information" OR "Mechanism of action information" OR "Drug pregnancy information" OR "Drug Lactation information" OR "Drug identification information" OR "Drug cost information" OR "Drug price information" OR "Drug therapy information" OR "dispensing information" OR "therapy information" OR "Pharmaceutic information" OR "Pharmaceutics information" OR "Pharmaceutical information" OR "Pharmacologic information" OR "Pharmacologics information" OR "Pharmacological information"):ti,ab	678
#9	MeSH descriptor: [Information Seeking Behavior] explode all trees	35
#10	MeSH descriptor: [Information Literacy] explode all trees	347
#11	(Information NEXT seeking NEXT behavio* OR Information NEXT seek* OR Information NEXT sourc* OR Information NEXT search* OR Information NEXT forag* OR Information NEXT retriev* OR Information NEXT gather* OR Information NEXT behavio* OR Information NEXT acqui* OR Information NEXT expect* OR "Seeking information" OR "Seek information" OR "Seek for information" OR "Seeking for information" OR "Gather information" OR "Seeking knowledge" OR "Seek knowledge" OR "Acquire knowledge"):ti,ab	1065
#12	#9 OR #10 OR #11	1428
#13	MeSH descriptor: [Needs Assessment] explode all trees	344
#14	(Information NEXT need* OR Information NEXT require* OR Information NEXT demand* OR Information NEXT concern* OR Information NEXT preference* OR information NEXT gap* OR Information NEXT wish* OR Information NEXT ask* OR Prescribing NEXT need* OR Knowledge NEXT need* OR Informational NEXT need*):ti,ab	1116
#15	#13 OR #14	1456
#16	MeSH descriptor: [Information Services] explode all trees	1857
#17	MeSH descriptor: [Drug Information Services] explode all trees	142
#18	(Information NEXT source* OR Information NEXT resource* OR Information NEXT reference* OR Information NEXT material* OR Information NEXT service* OR "Source of information" OR "Sources of information" OR "Source of drug information" OR "Sources of drug information" OR Drug NEXT reference* OR Drug NEXT resource* OR Drug NEXT database*):ti,ab	1075
#19	#16 OR #17 OR #18	3042
#20	(Information NEXT barrier* OR Information NEXT obstacle* OR Information NEXT challeng* OR Information NEXT facilitat* OR Information NEXT enabl*):ti,ab	57
#21	#12 OR #15 OR #19 OR #20	5758
#22	#7 AND #8 AND #21	29
#23	#7 AND #8 AND #21 (with COCHRANE LIBRARY publication date from Feb 2019 to May 2020)	9